DevOrSysAdminScripts

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

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Code lines: 0 total lines, 0 not empty lines, 0 empty lines.

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1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spell_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── DevOrSysAdminScripts.html.tpl
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_listing.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
│   ├── licenses_notes.txt
```

```
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── IMAP_transfert.py
├── LiensUtiles.txt
├── README.html
├── README.md
├── README.md.tpl
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

5 directories, 62 files

[ 12K Mar  2 19:33]  ./
├── [4.0K Mar  2 19:33]  build_and_checks_dependencies/
│   ├── [3.8K Mar  2 18:25]  build_and_checks.exec.sh.tpl*
│   ├── [3.8K Mar  2 19:19]  build_md_from_printable_md.exec.sh*
│   ├── [2.6K Mar  2 19:19]  check_shell_scripts_beginnings.libr.sh
│   ├── [4.5K Mar  2 19:19]  check_shell_scripts_indentation.libr.sh
│   ├── [3.2K Mar  2 19:19]  check_URLs.libr.sh
│   ├── [ 28K Mar  2 19:33]  common_build_and_checks.exec.sh*
│   ├── [ 23K Mar  2 19:20]  common_build_and_checks.exec.sh.tpl
│   ├── [6.0K Mar  2 19:32]  common_options.libr.sh
│   ├── [4.7K Feb 23 17:34]  comparisons.libr.sh
│   ├── [ 18K Mar  2 19:33]  create_PDF.exec.sh*
│   ├── [3.3K Feb 19 18:48]  generate_from_template.libr.sh
│   ├── [1.9K Feb 19 17:18]  get_common_text_glob_patterns.libr.sh
│   ├── [2.5K Mar  2 19:19]  grammar_and_spell_check.libr.sh
│   ├── [4.0K Mar  2 19:19]  licenses_templates/
│   │   ├── [ 13K Mar  2 19:19]  build_licenses_templates.exec.sh*
│   │   ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│   │   ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│   │   └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
```

```
│   ├── [3.9K Mar  2 19:19]  lines_counts.libr.sh
│   ├── [8.8K Feb 27 21:26]  lines_filters.libr.sh
│   ├── [2.7K Feb 19 16:16]  lines_maps.libr.sh
│   ├── [1.8K Feb 26 16:58]  overwrite_if_not_equal.libr.sh
│   ├── [2.2K Mar  2 19:19]  python_black_complement.libr.sh
│   ├── [2.5K Mar  2 19:19]  python_isort_complement.libr.sh
│   ├── [1.6K Feb  3 17:57]  split_score.exec.php
│   ├── [1.9K Feb  3 20:11]  split_score.libr.php
│   ├── [ 12K Feb 26 15:37]  strings_functions.libr.sh
│   ├── [3.5K Mar  2 19:19]  too_long_code_lines.libr.sh
│   ├── [4.1K Mar  1 22:38]  update_common_build_and_checks.exec.sh*
│   └── [5.1K Mar  2 19:19]  update_or_check_files_names_listing.exec.sh*
├── [4.0K Mar  2 19:33]  build_and_checks.exec.sh*
├── [4.0K Mar  2 19:33]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [ 16K Mar  2 12:57]  DevOrSysAdminScripts.html.tpl
│   ├── [ 28K Mar  2 19:32]  DevOrSysAdminScripts.tex
│   ├── [8.5K Mar  2 13:01]  DevOrSysAdminScripts.tex.tpl
│   ├── [ 13K Mar  2 15:42]  files_listing.tex.tpl
│   ├── [4.1K Mar  1 22:37]  files_names_listing.txt
│   ├── [  24 Jan 15 22:26]  .gitignore
│   ├── [4.9K Mar  2 12:48]  licenses_notes.txt
│   ├── [1001 Feb 26 15:37]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.4K Feb 28 20:28]  repository_data.txt
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 167 Feb 19 18:39]  shellcheck.ini
│   └── [2.1K Feb 19 19:36]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [353K Mar  2 19:32]  DevOrSysAdminScripts.html
├── [323K Mar  2 19:33]  DevOrSysAdminScripts.pdf
├── [4.0K Feb 27 20:20]  git_helpers/
│   ├── [3.5K Feb 19 17:00]  git_checks.libr.sh
│   ├── [1.8K Feb 27 20:20]  post-commit
│   ├── [ 13K Feb 26 19:35]  post_commit_renaming_comment.libr.sh
│   └── [1.4K Feb 19 18:42]  pre-commit
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [ 830 Dec 29 19:10]  LiensUtiles.txt
├── [7.4K Mar  1 22:30]  README.html
├── [3.7K Mar  1 22:30]  README.md
├── [3.7K Mar  1 22:29]  README.md.tpl
├── [9.1K Feb 26 15:05]  unicode.libr.php
├── [1.4K Feb  3 14:19]  upgrade_build_and_checks.exec.sh*
```

```
└── [1.4K Feb 19 14:41]  wget_sha512.libr.sh

5 directories, 62 files
```

2 Listing of files

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
  basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
https://www.gnu.org/software/bash/,
https://tiswww.case.edu/php/chet/bash/bashtop.html,
https://savannah.gnu.org/projects/bash/.

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/,
https://savannah.gnu.org/projects/coreutils/.

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/,
https://github.com/uutils/coreutils.

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/,
https://savannah.gnu.org/projects/diffutils/.

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/,
https://savannah.gnu.org/projects/findutils/.

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/,
https://savannah.gnu.org/projects/grep/.

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/,
https://savannah.gnu.org/projects/sed/.

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/,
https://github.com/koalaman/shellcheck.

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/,
https://savannah.gnu.org/projects/wget/.

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/,
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/,
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,
https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

If you see something that I missed regarding my dependencies, please tell me/email me :).

2.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

```

```
  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    {
      echo "temp/"
      echo ""
    } > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  @sha512_common_build_and_checks.exec.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}

if [[ "$*" == *--fixed_point_build* ]]; then
  echo "--fixed_point_build"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
```

```
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed_point_build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.2 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "build_md_from_printable_md.sh"
```

```
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  local LFBFL_base_name="README"
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --base-name=* ]]; then
      LFBFL_base_name=${LFBFL_arg#--base-name=}
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        echo "Searching md file: ${LFBFL_base_name}"
      fi
      break
    fi
  done
  readonly LFBFL_base_name

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]];
```

```
  then
    sed -Ez -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
      "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    echo "No file ${LFBFL_base_name}.md.tpl"
  fi

  pandoc --from markdown --to html --standalone\
    --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
    -o "${LFBFL_base_name}.html.temp"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed -i -E -e 's~(  <meta .*) />~\1>~g'\
            -e 's~<html .*>~<html lang="en">~'\
            -e '/    code\{white-space: pre-wrap;\}/d'\
    "${LFBFL_base_name}.html.temp"
  sed -i -Ez -e 's~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
             -e "${LFBFL_prewrap}"\
    "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

2.3 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
```

```
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# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # declare -r LFBFL_file_name=$(basename "$1")
  # if [[ "${LFBFL_file_name}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head -n 1 "$1") <(echo '#!/usr/bin/env bash')
  # ---------------------------------------------------------------
  # is of course slower than
  # ---------------------------------------------------------------
  # diff <(head -n 1 "$1") <(echo '#!/usr/bin/env bash')
  # ---------------------------------------------------------------
  # is of course slower than
  # ---------------------------------------------------------------
  declare -r LFBFL_head=$(head -n 1 "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || echo "$1:File has wrong shell script beginning."
}
```

```

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  shopt -s globstar
  local LFBFL_file_name
  local LFBFL_head

  for LFBFL_file_name in **/*.sh; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && echo "${LFBFL_file_name}:Checking shell script beginning."
    LFBFL_head=$(head -n 1 "${LFBFL_file_name}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || echo "${LFBFL_file_name}:File has wrong shell script beginning."
  done
}
```

2.4 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
#!/usr/bin/env bash
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```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_one_shell_script_indentation(){
  shopt -s lastpipe

  declare -i LFBFL_file_with_error=0
  local LFBFL_some_line
  local LFBFL_previous_line="workeduntilsomeasshole"
  # created a command named "workeduntilsomeasshole"
  # and called it without neither indentation, nor parameters
  # in a .sh script XD.
  # Manual justification of the lines L-4 and L-3 was a serendipity.
  # I just tried a joke and was lucky ;) :).
  local LFBFL_line_end
  local LFBFL_previous_line_end
  declare -i LFBFL_line_length
  declare -i LFBFL_line_end_length
  declare -i LFBFL_line_start_length
  local LFBFL_spacing
  declare -i LFBFL_offset
  local LFBFL_substring1
  local LFBFL_substring2
  declare -r LFBFL_pattern='^ *-e '
  # This grep would have been enough without some sed commands.
  # shellcheck disable=SC2312
  grep -EHn -B 1 '^(  )* ([^ ]|$)' "$1"\
    | while read -r LFBFL_some_line;
  do
    if [[ "${LFBFL_some_line}" == "--" ]]; then
      # When switching between code fragments, we reinit.
      LFBFL_previous_line="workeduntilsomeasshole"
      continue
    fi
    # Note how it's funny that this loop always work because we check
    # shell scripts beginning in another script and thus we always
    # have a line before.
    if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]];
```

```
    then
      LFBFL_previous_line="${LFBFL_some_line}"
      continue
    fi
    # Remove the name of the file and then the line number
    LFBFL_line_end=${LFBFL_some_line#*:}
    LFBFL_line_end=${LFBFL_line_end#*:}
    # But for the previous line '-' or ':' is used instead of ':'
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_line_length=${#LFBFL_some_line}
    LFBFL_line_end_length=${#LFBFL_line_end}
    LFBFL_line_start_length=$((
      LFBFL_line_length-LFBFL_line_end_length
    ))
    LFBFL_offset=$((LFBFL_line_start_length-2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_offset+=1));;
      :) ((LFBFL_offset+=1));;
      [0-9]) ((LFBFL_offset+=2));;
      *) echo ':-(';;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        echo "${LFBFL_some_line}"
        LFBFL_file_with_error=1
      fi
    else
      echo "${LFBFL_some_line}"
      LFBFL_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_file_with_error -eq 1 ]]; then
    echo "$1:File has some lines with odd number of spaces."
```

```
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  shopt -s globstar
  local LFBFL_file_name

  for LFBFL_file_name in **/*.sh; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && echo "${LFBFL_file_name}:Checking shell script indentation."
    check_one_shell_script_indentation "${LFBFL_file_name}"
  done
}
```

2.5 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
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```
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#
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# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && echo "pipefail option activated"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )

  local LFBFL_pattern
  local LFBFL_file_name
  local LFBFL_base_file_name
  local LFBFL_substitution
  local LFBFL_substitution2
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 0 ]]\
      || echo "Iterating on pattern: ${LFBFL_pattern}"
```

```
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H 'http:'\
      | grep -v "| xargs grep -H 'htt"\
      | grep -vP "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"
    # Last grep just above will remove false positives from
    # substitutions that fit on one line.
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find\
      | while read -r LFBFL_file_name;
    do
      [[ -f "${LFBFL_file_name}" ]] || continue
      LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      [[ LFBFL_i_verbose -eq 0 ]]\
        || echo "Handling the file: ${LFBFL_file_name}"
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=\
${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep -q "${LFBFL_substitution}" "${LFBFL_file_name}"; then
          sed -i "s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            "${LFBFL_file_name}"
        fi
      done
    done
  done
}
```

2.6 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  # declare -r LFBFL_dependencies_URL="$2" too long
  declare -r LFBFL_start_URL="$2"
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  if [[ ! -o pipefail ]]; then
    [[ "${LFBFL_verbose}" == "--verbose" ]]\
      && echo "pipefail option activated"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  local LFBFL_file_name
  local LFBFL_URL
  local LFBFL_file_path

```

```
LFBFL_file_name="build_md_from_printable_md.exec.sh"
# LFBFL_script_download_URL
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='debb2bdda6f77ef82be5c17a547077d5c3db2bea62f92'
LFBFL_correct_sha512+='6b092815c4d7a2dbd9b8461923bff8e38199d776a71b'
LFBFL_correct_sha512+='c8d417fa396a556898c3d783681afb8c235922f'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='239c41539d866e50bfbe32e91b4ebaf80e751e2621e86'
LFBFL_correct_sha512+='4c70c01b918f1d563eba5cca49bca8967c292edc9c29'
LFBFL_correct_sha512+='1fd372c53cad764b9bfb742b5dbdc0d4832db5c'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='4afb238c86c2f4ea796364a2c4462f113396d6f60ace1'
LFBFL_correct_sha512+='b8602fa56ddae83645f541bbf6be0d200264c36b8cab'
LFBFL_correct_sha512+='6c74a75807cb09d5c1712f3d810dc0e8c49dc0b'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2c29d1fda4042636e7302f72de571f92d4f7e470310bf'
LFBFL_correct_sha512+='1180af796e5d710cd008b504d755d0951cb3f76bc63f'
LFBFL_correct_sha512+='dcd2dcb480f2baa16637101ca4216a2be93f54d'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="common_options.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
```

```
LFBFL_correct_sha512='71329222e7e691f1e93de4ce98e3f15d7f4344a3dd244'
LFBFL_correct_sha512+='be0d82188b1fac45e4a1c9853a5ac8c757c10dc798f2'
LFBFL_correct_sha512+='d1c3dc5567b9f99e6a72f3bdd15e5c2061cf47d'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="comparisons.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='1abcc547d6baf661c339489967108fd744321419e8e6d'
LFBFL_correct_sha512+='2a25896280a23243106dc72be30a4f400caf6789e9be'
LFBFL_correct_sha512+='740223b9e12d6155bb931aaf94bb946455c8c5c'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='586b6b9bbbae9c17fe0e7a9684376100ff5a08488aba8'
LFBFL_correct_sha512+='412278d6cbe48002906e57e554c702cb38fa463b2694'
LFBFL_correct_sha512+='d3aa277eb882a59810b54e70ba6185f79c6c53e'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='5779c3d2c64d0bb403525c410e37b0b06fcf1ef97d503'
LFBFL_correct_sha512+='21fd27345545a8ebea020b2d3d809ac5c1bc66f9cebe'
LFBFL_correct_sha512+='a8e8ea63aa539eaa179cf09a3860239c96df8e9'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='4e0915efcc9a4088b1f79c9c7efbf7da5d68e8e1b21ff'
LFBFL_correct_sha512+='feb4746c9d468be80affa30babf35938e15f9161db3e'
LFBFL_correct_sha512+='3f6d95c913b86638563d7bd2995dc58a26f8e3a'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
```

```

LFBFL_file_name="grammar_and_spell_check.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='90d133adf3dc9a7b75f18e5ab64846157e2b36a1f83e7'
LFBFL_correct_sha512+='1bc1f0a31781c978167593f51d6f9775f5c4091df586'
LFBFL_correct_sha512+='7c7cfda01ebe11475eb9a71ddc3b365cc345d3f'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

# /licenses_templates/ ---------------------------------------------
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='7cae7c2e3cd7c3146f1c8752920cc6be238227b12e32b'
LFBFL_correct_sha512+='4fdb3061126ecb8e2e077882ef8ecf10f248bca1c43c'
LFBFL_correct_sha512+='f18810ac71bd6ae3f152a25ab7aff7723d7dc8f'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
```

```
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='f13bad4bd7091d8719706be7cf5df5fa024899c43df6d'
LFBFL_correct_sha512+='38317bd46a893d059a82cb3481727648862713ba844d'
LFBFL_correct_sha512+='4d2ff8841954ecbc2c1fe7297914754c1e98cbd'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='1a5a7a4903168a61135b4cbd1d6ac65c65a93b6873130'
LFBFL_correct_sha512+='8115edfb0ff4dfe64321c7b6b1fe15a1a32aff292736'
LFBFL_correct_sha512+='a74645974ca01f0162e5e9f753910281e3576f0'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='25c3e5082951260295b5f0e87ba1f2c978efdea1e8926'
LFBFL_correct_sha512+='072ce4d2062dc2a47bb92b9098f081e553eeae5d9154'
LFBFL_correct_sha512+='fdc50c9d88ae5365b776a23eb6e026b884a8374'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2781a564d264bae84c787321481dc83ae43a1ce324ee6'
LFBFL_correct_sha512+='d5ceb73d5a98607fad5fe9ff8d03979b898f7cd7c712'
```

```
LFBFL_correct_sha512+='8215d26e76874d56aa67fb63fa9de776617d24a'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='38058e326270c15885c5b01112c984fbef97bcf82c7b7'
LFBFL_correct_sha512+='a8023a496edb38c187808360170861e27c8e6a770d44'
LFBFL_correct_sha512+='a4f265732ac969c2dd875fcf383d081a35c029f'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='d8a40cf571c4140d8565e1e1e752a99ce68e9d514725d'
LFBFL_correct_sha512+='194236a000d2127330d53e6be70e3312a2d61bff4542'
LFBFL_correct_sha512+='67279a31e22ab1bade53b8be9644118e9e4ea21'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.exec.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2f7ff107a43716109b1acbd7d89870d05577af00a981a'
LFBFL_correct_sha512+='05f162a054c96387fb99544f8225c333b1212f74326c'
LFBFL_correct_sha512+='f9ec6196e796f6257dfbba9347ac8c57b4f09db'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.libr.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='1413898a4fb4f31ca17993fd6110f7e1f32adf31c22c4'
LFBFL_correct_sha512+='7fa46a31d3c1511a3112a611e04fac19bb838d4476d9'
LFBFL_correct_sha512+='ecf77a28dd48d3d2732df7392c50e38262852d7'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='f29cce1e27d2b9483716326938b09b4e9cffdc23f5a39'
LFBFL_correct_sha512+='1516dd6b1514903b3484b38e9c585c85ff389e5d90a0'
LFBFL_correct_sha512+='2ffda748385aac01a078dc8aae06f78c08579b6'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='692002fd362636fbb0ed8fd5345c6325d43b87b262a2a'
LFBFL_correct_sha512+='1d258661221ddc6a7324d27e7a9dbe2149fa291fcb46'
LFBFL_correct_sha512+='d8e33bf9df03d71b7d3bda4f827250bd2ca7991'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='78181135c4660fd94a130caf3637100c416bf136c678e'
LFBFL_correct_sha512+='4431f76e66217cd3967cbb419890e4ea60eeeaed435f'
LFBFL_correct_sha512+='fce28e16c9153e04666237e1eeafbb2eaf2f25e'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

shopt -s globstar
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spell_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
```

```
  # shellcheck source=lines_counts.libr.sh
  source "./${LFBFL_subdir}/lines_counts.libr.sh"
  # shellcheck source=lines_filters.libr.sh
  source "./${LFBFL_subdir}/lines_filters.libr.sh"
  # shellcheck source=lines_maps.libr.sh
  source "./${LFBFL_subdir}/lines_maps.libr.sh"
  # shellcheck source=overwrite_if_not_equal.libr.sh
  source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
  # shellcheck source=python_black_complement.libr.sh
  source "./${LFBFL_subdir}/python_black_complement.libr.sh"
  # shellcheck source=python_isort_complement.libr.sh
  source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
  # shellcheck source=strings_functions.libr.sh
  source "./${LFBFL_subdir}/strings_functions.libr.sh"
  # shellcheck source=too_long_code_lines.libr.sh
  source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

  declare -i LFBFL_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_upgrade_venvs_ts
  declare -i LFBFL_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"
  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
```

```
    *) echo "No wat for you?";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      LFBFL_upgrade_venvs=1
    fi
  else
    LFBFL_upgrade_venvs=1
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_upgrade_venvs=0
      fi
    fi
  fi

  echo "Building license headers"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh" "$@"

  echo "Building README.md"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_work_directory}"\
    "--base-name=README"\
    "${LFBFL_verbose}"

  echo "Building other MarkDown files"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  find "${LFBFL_work_directory}" -name "*.md.tpl"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
```

```
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    echo "Found template ${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
    LFBFL_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_file_name=${LFBFL_file_name%.md.tpl}
    "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
      "--work-directory=${LFBFL_some_directory}"\
      "--base-name=${LFBFL_file_name}"\
      "${LFBFL_verbose}"
  done

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  echo "Running shellcheck"
  find . -name "*.sh"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
      "${LFBFL_file_path}"
  done

  echo "---Python---"
  echo "Running isort"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  echo "Running python_isort_complement"
```

```
  python_isort_complement "$@"

  echo "Running black"
  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  echo "Running python_black_complement"
  python_black_complement "$@"

  echo "Probing if mypy should be runned"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_no_toml=1
  find . -name "pyproject.toml"\
    | relevant_find\
```

```
    | while read -r LFBFL_file_path;
  do
    if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
      echo "Running mypy"
      LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
      mypy "${LFBFL_directory_path}"
    fi
    LFBFL_no_toml=0
  done
  if [[ LFBFL_no_toml -eq 1 ]]; then
    echo "Running mypy"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  echo "Running bandit"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini build_and_checks_variables/bandit.ini\
    -b build_and_checks_variables/bandit_baseline.json\
    -r .
  # Saving new baseline in temp if necessary.
  bandit --ini build_and_checks_variables/bandit.ini\
    -f json -o build_and_checks_variables/temp/bandit_baseline.json\
    -r .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  echo "Running pylint"
  LFBFL_pylint_venv=""
```

```
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile build_and_checks_variables/pylintrc.toml\
    --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  echo "Running ruff"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config build_and_checks_variables/ruff.toml
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  echo "---Python end---"

  echo "---PHP---"
  echo "Running PHPMD"
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
```

```
614     fi
615
616     local LFBFL_phpmd_baseline="build_and_checks_variables/"
617     LFBFL_phpmd_baseline+="phpmd_baseline.xml"
618     readonly LFBFL_phpmd_baseline
619
620     local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
621     LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
622     readonly LFBFL_temp_phpmd_baseline
623
624     local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
625     LFBFL_phpmd_rulesets+="design,naming,unusedcode"
626     readonly LFBFL_phpmd_rulesets
627
628     phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
629       . text "${LFBFL_phpmd_rulesets}"
630     # Saving new baseline in temp if necessary.
631     rm "${LFBFL_temp_phpmd_baseline}"
632     phpmd --color --generate-baseline\
633       --baseline-file "${LFBFL_temp_phpmd_baseline}"\
634       . text "${LFBFL_phpmd_rulesets}"
635     echo "---PHP end---"
636
637     echo "---JS---"
638     LFBFL_npm_lint_directories=""
639     grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
640       --result-variable-prefix="LFBFL_"
641     if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
642       echo "Running ESLint"
643       local LFBFL_JS_directory
644       echo "${LFBFL_npm_lint_directories}"\
645         | sed -e 's/\\n/\n/g'\
646         | while read -r LFBFL_JS_directory;
647       do
648         (cd "${LFBFL_JS_directory}" && npm run lint)
649       done
650     fi
651     echo "---JS end---"
652
653     echo "Analyzing too long lines"
654     # shellcheck disable=SC2248
655     too_long_code_lines "${LFBFL_verbose}"\
656       --max-line-length=${LFBFL_max_line_length}
657
658     echo "Analyzing shell scripts beginnings"
659     check_shell_scripts_beginnings "${LFBFL_verbose}"\
```

```
    | relevant_grep

  echo "Analyzing shell scripts indentation"
  check_shell_scripts_indentation "${LFBFL_verbose}"\
    | relevant_grep

  echo "Analyzing URLs"
  check_URLs "$@" \
    | relevant_grep

  echo "Analyzing strange characters: hover over in doubt"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  echo "Checking listed files"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"

  echo "Creating the PDF file of the listing of the source code"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_verbose}"

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  echo "Running Nu W3C HTML CSS and SVG validator"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
```

```
    LFBFL_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_vnu_jar_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      echo $'/!\\ATTENTION: Nu W3C validator is too old./!\\'
      echo "Go get the latest version here:"
      echo "https://github.com/validator/validator/releases"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    declare -a LFBFL_files_for_Nu2
    mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
    readonly LFBFL_files_for_Nu2
    java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
      --verbose "${LFBFL_files_for_Nu2[@]}"
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.7 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
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# of the GNU General Public License
# as published by the Free Software Foundation,
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# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
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```

```
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  # declare -r LFBFL_dependencies_URL="$2" too long
  declare -r LFBFL_start_URL="$2"
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  if [[ ! -o pipefail ]]; then
    [[ "${LFBFL_verbose}" == "--verbose" ]]\
      && echo "pipefail option activated"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  local LFBFL_file_name
  local LFBFL_URL
  local LFBFL_file_path

  LFBFL_file_name="build_md_from_printable_md.exec.sh"
```

```
# LFBFL_script_download_URL
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_md_from_printable_md.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_beginnings.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_indentation.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_URLs.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="common_options.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_common_options.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="comparisons.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_comparisons.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
@sha512_create_PDF.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_generate_from_template.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_get_common_text_glob_patterns.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="grammar_and_spell_check.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_grammar_and_spell_check.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

# /licenses_templates/ ---------------------------------------------
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_build_licenses_templates.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_AGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_GPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_LGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_counts.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_filters.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_maps.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_overwrite_if_not_equal.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_black_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
```

```
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="python_isort_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_python_isort_complement.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="split_score.exec.php"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_split_score.exec.php@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="split_score.libr.php"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_split_score.libr.php@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="strings_functions.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_strings_functions.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="too_long_code_lines.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_too_long_code_lines.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_update_or_check_files_names_listing.exec.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  shopt -s globstar
```

```
  # shellcheck source=check_shell_scripts_beginnings.libr.sh
  source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
  # shellcheck source=check_shell_scripts_indentation.libr.sh
  source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
  # shellcheck source=check_URLs.libr.sh
  source "./${LFBFL_subdir}/check_URLs.libr.sh"
  # shellcheck source=common_options.libr.sh
  source "./${LFBFL_subdir}/common_options.libr.sh"
  # shellcheck source=comparisons.libr.sh
  source "./${LFBFL_subdir}/comparisons.libr.sh"
  # shellcheck source=generate_from_template.libr.sh
  source "./${LFBFL_subdir}/generate_from_template.libr.sh"
  # shellcheck source=get_common_text_glob_patterns.libr.sh
  source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
  # shellcheck source=grammar_and_spell_check.libr.sh
  source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
  # shellcheck source=lines_counts.libr.sh
  source "./${LFBFL_subdir}/lines_counts.libr.sh"
  # shellcheck source=lines_filters.libr.sh
  source "./${LFBFL_subdir}/lines_filters.libr.sh"
  # shellcheck source=lines_maps.libr.sh
  source "./${LFBFL_subdir}/lines_maps.libr.sh"
  # shellcheck source=overwrite_if_not_equal.libr.sh
  source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
  # shellcheck source=python_black_complement.libr.sh
  source "./${LFBFL_subdir}/python_black_complement.libr.sh"
  # shellcheck source=python_isort_complement.libr.sh
  source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
  # shellcheck source=strings_functions.libr.sh
  source "./${LFBFL_subdir}/strings_functions.libr.sh"
  # shellcheck source=too_long_code_lines.libr.sh
  source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
```

```
    build_and_checks_variables/black.toml.temp

  declare -i LFBFL_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_upgrade_venvs_ts
  declare -i LFBFL_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"
  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) echo "No wat for you?";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      LFBFL_upgrade_venvs=1
    fi
  else
    LFBFL_upgrade_venvs=1
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_upgrade_venvs=0
      fi
    fi
  fi

```

```
  echo "Building license headers"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh" "$@"

  echo "Building README.md"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_work_directory}"\
    "--base-name=README"\
    "${LFBFL_verbose}"

  echo "Building other MarkDown files"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  find "${LFBFL_work_directory}" -name "*.md.tpl"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    echo "Found template ${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
    LFBFL_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_file_name=${LFBFL_file_name%.md.tpl}
    "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
      "--work-directory=${LFBFL_some_directory}"\
      "--base-name=${LFBFL_file_name}"\
      "${LFBFL_verbose}"
  done

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  echo "Running shellcheck"
  find . -name "*.sh"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
      "${LFBFL_file_path}"
  done

  echo "---Python---"
  echo "Running isort"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
```

```
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  echo "Running python_isort_complement"
  python_isort_complement "$@"

  echo "Running black"
  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  echo "Running python_black_complement"
  python_black_complement "$@"

  echo "Probing if mypy should be runned"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
```

```
if [[ -n "${LFBFL_mypy_venv}" ]]; then
  if command -v deactivate
  then
    deactivate
  fi
  # shellcheck disable=SC1090,SC1091
  source "${LFBFL_mypy_venv}/bin/activate"
fi
if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade mypy
fi
shopt -s lastpipe
local LFBFL_directory_path
declare -i LFBFL_no_toml=1
find . -name "pyproject.toml"\
  | relevant_find\
  | while read -r LFBFL_file_path;
do
  if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
    echo "Running mypy"
    LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
    mypy "${LFBFL_directory_path}"
  fi
  LFBFL_no_toml=0
done
if [[ LFBFL_no_toml -eq 1 ]]; then
  echo "Running mypy"
  mypy .
fi
if [[ -n "${LFBFL_mypy_venv}" ]]; then
  deactivate
fi

echo "Running bandit"
LFBFL_bandit_venv=""
grep_variable "${LFBFL_data_file_name}" bandit_venv\
  --result-variable-prefix="LFBFL_"\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_bandit_venv}" ]]; then
  if command -v deactivate
  then
    deactivate
  fi
  # shellcheck disable=SC1090,SC1091
  source "${LFBFL_bandit_venv}/bin/activate"
fi
```

```
477   if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
478     pip install --upgrade bandit
479   fi
480   bandit --ini build_and_checks_variables/bandit.ini\
481     -b build_and_checks_variables/bandit_baseline.json\
482     -r .
483   # Saving new baseline in temp if necessary.
484   bandit --ini build_and_checks_variables/bandit.ini\
485     -f json -o build_and_checks_variables/temp/bandit_baseline.json\
486     -r .
487   if [[ -n "${LFBFL_bandit_venv}" ]]; then
488     deactivate
489   fi
490
491   echo "Running pylint"
492   LFBFL_pylint_venv=""
493   grep_variable "${LFBFL_data_file_name}" pylint_venv\
494     --result-variable-prefix="LFBFL_"\
495     --replace-line-returns-by=""
496   if [[ -n "${LFBFL_pylint_venv}" ]]; then
497     if command -v deactivate
498     then
499       deactivate
500     fi
501     # shellcheck disable=SC1090,SC1091
502     source "${LFBFL_pylint_venv}/bin/activate"
503   fi
504   if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
505     pip install --upgrade pylint
506   fi
507   pylint --rcfile build_and_checks_variables/pylintrc.toml\
508     --recursive=y .
509   if [[ -n "${LFBFL_pylint_venv}" ]]; then
510     deactivate
511   fi
512
513   echo "Running ruff"
514   LFBFL_ruff_venv=""
515   grep_variable "${LFBFL_data_file_name}" ruff_venv\
516     --result-variable-prefix="LFBFL_"\
517     --replace-line-returns-by=""
518   if [[ -n "${LFBFL_ruff_venv}" ]]; then
519     if command -v deactivate
520     then
521       deactivate
522     fi
```

```
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config build_and_checks_variables/ruff.toml
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  echo "---Python end---"

  echo "---PHP---"
  echo "Running PHPMD"
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  local LFBFL_phpmd_baseline="build_and_checks_variables/"
  LFBFL_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_phpmd_baseline

  local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

  local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
  LFBFL_phpmd_rulesets+="design,naming,unusedcode"
  readonly LFBFL_phpmd_rulesets

  phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  # Saving new baseline in temp if necessary.
  rm "${LFBFL_temp_phpmd_baseline}"
  phpmd --color --generate-baseline\
    --baseline-file "${LFBFL_temp_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  echo "---PHP end---"

  echo "---JS---"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
    --result-variable-prefix="LFBFL_"
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    echo "Running ESLint"
    local LFBFL_JS_directory
```

```
    echo "${LFBFL_npm_lint_directories}"\
      | sed -e 's/\\n/\n/g'\
      | while read -r LFBFL_JS_directory;
    do
      (cd "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  echo "---JS end---"

  echo "Analyzing too long lines"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_verbose}"\
    --max-line-length=${LFBFL_max_line_length}

  echo "Analyzing shell scripts beginnings"
  check_shell_scripts_beginnings "${LFBFL_verbose}"\
    | relevant_grep

  echo "Analyzing shell scripts indentation"
  check_shell_scripts_indentation "${LFBFL_verbose}"\
    | relevant_grep

  echo "Analyzing URLs"
  check_URLs "$@" \
    | relevant_grep

  echo "Analyzing strange characters: hover over in doubt"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©"”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  echo "Checking listed files"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"

  echo "Creating the PDF file of the listing of the source code"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_verbose}"

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  echo "Running Nu W3C HTML CSS and SVG validator"
```

```
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_vnu_jar_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      echo $'/!\\ATTENTION: Nu W3C validator is too old./!\\'
      echo "Go get the latest version here:"
      echo "https://github.com/validator/validator/releases"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    declare -a LFBFL_files_for_Nu2
    mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
    readonly LFBFL_files_for_Nu2
    java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
      --verbose "${LFBFL_files_for_Nu2[@]}"
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.8 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

is_top_dirstack_directory(){
  # This function should not be needed since when testing "${DIRSTACK[0]}"
  # is already a realpath. But I keep it, in case it is needed.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
```

```
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd requested: $1"
  declare -ig enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  fi
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd validated1"
  is_top_dirstack_directory "$1" && {
    enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd validated2"

  pushd "$1" || {
    enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    echo "${LFBFL_where_was_i} $3$1 no such directory."
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd executed"
  return 0
}

enhanced_popd(){
  # Auxiliary function needed below.
```

```
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_popd requested: $1"
  if [[ -z "$1" ]]; then
    return 0
  fi

  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    echo "${LFBFL_where_was_i} popd failed."
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_popd executed"
  return 0
}

can_continue_after_enhanced_pushd(){
  [[ enhanced_pushd_result -eq 0\
  || enhanced_pushd_result -eq 110\
  || enhanced_pushd_result -eq 111\
  ]]
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
```

```
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  # This test is faster than:
  #   for LFBFL_arg in "$@"; do
  #     if [[ "${LFBFL_arg}" == "--verbose" ]]; then
  # But it is not bug free.
  if [[ "$*" == *--verbose* ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    readonly LFBFL_i_popd_result
    echo "${LFBFL_where_was_i} $*"
    LFBFL_i_verbose=1
  fi
  # shellcheck disable=SC2034
  readonly LFBFL_i_verbose
}


# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --work-directory=* ]]; then
      LFBFL_work_directory=${LFBFL_arg#--work-directory=}
      LFBFL_work_directory=${LFBFL_work_directory:-.}
      # LFBFL_work_directory=${LFBFL_work_directory/#~/${HOME}} line below
```

```
      LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
      break
    fi
  done
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
}

popd_from_work_directory(){
  enhanced_popd "${LFBFL_work_directory}" 3
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}
```

2.9 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

```

```
26 equal(){
27   # Returns 1 if all the arguments strings are equal.
28   # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
29   # J'ai rerepensé au 3-way-merge
30   # et à mon article sur le principe de première différence
31   # appliqué à la largeur modulaire, de clique ou de rang avec
32   # l'idée d'un principe de première différence ternaire.
33   declare -i LFBFL_first=1
34   local LFBFL_current
35   local LFBFL_arg
36   for LFBFL_arg in "$@"; do
37     if [[ LFBFL_first -eq 1 ]]; then
38       LFBFL_current="${LFBFL_arg}"
39       LFBFL_first=0
40       continue
41     fi
42     if [[ "${LFBFL_current}" != "${LFBFL_arg}" ]]; then
43       return 0
44     fi
45   done
46   return 1
47 }
48
49 some_distinct(){
50   ! equal "$@"
51 }
52
53 all_distinct(){
54   # Returns 1 if all the arguments strings are distinct.
55   declare -ir LFBFL_ij_max=$#
56   declare -i LFBFL_i
57   declare -i LFBFL_j
58   for ((LFBFL_i=1; LFBFL_i<=LFBFL_ij_max; ++LFBFL_i)); do
59     for ((LFBFL_j=1; LFBFL_j<=LFBFL_ij_max; ++LFBFL_j)); do
60       [[ LFBFL_i -eq LFBFL_j ]] && continue
61       [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
62     done
63   done
64   return 1
65 }
66
67 all_distinct2(){
68   # Returns 1 if all the arguments strings are distinct.
69   local LFBFL_element
70   local LFBFL_previous_element
71   if [[ $# -eq 0 ]]; then
```

```
    return 1
  fi
  # Not setting pipefail since the result would still be incorrect.
  # shellcheck disable=SC2312
  printf "%s\n" "$@" \
    | sort\
    | {
      read -r LFBFL_previous_element\
      && while read -r LFBFL_element; do
        if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]];
        then
          return 0
        fi
        LFBFL_previous_element="${LFBFL_element}"
      done
      return 1
    }
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # This is the fastest of all 3; but note that version 2 is faster
  # than this one if a call to sort is added.
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  # Not setting pipefail since the result would still be incorrect.
  # shellcheck disable=SC2162,SC2312
  printf "%s\n" "$@" \
    | uniq -d\
    | read
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
}

max(){
```

```
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
  # $1=$sort command with the flags you want to be applied.
  # $ max sort a b c
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
  printf "%s\n" "${@:2}" \
    | eval "$1 -r"\
    | head -1
}

min(){
  # Returns the minimum string among the arguments.
  # Renvoie la chaîne la plus petite parmi celles proposées.
  # $1=$sort command with the flags you want to be applied.
  # $ min sort a b c
  # a
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
  printf "%s\n" "${@:2}" \
    | eval "$1"\
    | head -1
}

is_substring(){
  # $1=$string
  # $2=$substring
  declare -r LFBFL_var_1=$(
    echo "$1"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\&/g"
  )
  declare -r LFBFL_var_2=$(
    echo "$2"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\&/g"
  )
```

```
  local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php -r "${LFBFL_var_3}"
  return $?
}

is_subfile(){
  # $1=$file_name
  # $2=$sub_file_name
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php -r "${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

2.10 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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#
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# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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#
# You should have received a copy of
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# along with DevOrSysAdminScripts.
```

```
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && echo "pipefail option activated"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  local LFBFL_subdir2="build_and_checks_variables"
  local LFBFL_data_file_name="${LFBFL_subdir2}/repository_data.txt"
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

```

```
  local LFBFL_repository_web_url=""
  grep_variable "${LFBFL_data_file_name}" repository_web_url\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_repository_git_url=""
  grep_variable "${LFBFL_data_file_name}" repository_git_url\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_abstract=""
  grep_variable "${LFBFL_data_file_name}" abstract\
    --result-variable-prefix="LFBFL_"

  local LFBFL_acknowledgments=""
  grep_variable "${LFBFL_data_file_name}" acknowledgments\
    --result-variable-prefix="LFBFL_"

  local LFBFL_author_email=""
  grep_variable "${LFBFL_data_file_name}" author_email\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_name}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "

  local LFBFL_author_website=""
  grep_variable "${LFBFL_data_file_name}" author_website\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  declare -r LFBFL_current_date=$(date -I"date")

  declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)

  declare -ir LFBFL_number_of_commits=$(
    git shortlog\
    | space_starting_lines\
    | wc -l
  )
```

```

  declare -ir LFBFL_all_code_lines_count=$(code_lines_count_all "$@")
  declare -ir LFBFL_not_empty_code_lines_count=$(
    code_lines_count_not_empty "$@"
  )
  declare -ir LFBFL_empty_code_lines_count=$(
    code_lines_count_empty "$@"
  )

  local LFBFL_number_of_lines="${LFBFL_all_code_lines_count}"
  LFBFL_number_of_lines+=" total lines,"
  LFBFL_number_of_lines+=" ${LFBFL_not_empty_code_lines_count}"
  LFBFL_number_of_lines+=" not empty lines,"
  LFBFL_number_of_lines+=" ${LFBFL_empty_code_lines_count}"
  LFBFL_number_of_lines+=" empty lines."
  readonly LFBFL_number_of_lines

  # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
  # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
  # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
  tree -a --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_subdir2}/temp/current_tree_light.txt"

  tree -a -DFh --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_subdir2}/temp/current_tree.txt.temp"

  local LFBFL_temp_files_listing="./${LFBFL_subdir2}/temp/"
  LFBFL_temp_files_listing+="files_listing.tex.tpl.temp"
  readonly LFBFL_temp_files_listing
  : > "${LFBFL_temp_files_listing}"
  local LFBFL_temp_files_listing2="./${LFBFL_subdir2}/temp/"
  LFBFL_temp_files_listing2+="files_listing.html.tpl.temp"
  readonly LFBFL_temp_files_listing2
  : > "${LFBFL_temp_files_listing2}"
```

```
  # HTML <li> elements, hence "lis".
  local LFBFL_temp_files_lis
  LFBFL_temp_files_lis="./${LFBFL_subdir2}/temp/files_lis.html.tpl"
  readonly LFBFL_temp_files_lis
  : > "${LFBFL_temp_files_lis}"
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_max_line_length} /
  declare -r LFBFL_score_command="${get_split_score_result}"
  local LFBFL_score_command_properties="${get_split_score_result2}"
  readonly LFBFL_score_command_properties
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_max_line_length} ':'
  declare -r LFBFL_score_command2="${get_split_score_result}"
  local LFBFL_score_command_properties2="${get_split_score_result2}"
  readonly LFBFL_score_command_properties2
  declare -r LFBFL_suffix='%'
  declare -i LFBFL_i=0
  local LFBFL_cleaned_path1
  local LFBFL_cleaned_path2
  local LFBFL_new_lines
  local LFBFL_new_lines2
  local LFBFL_new_lines3
  local LFBFL_file_name
  # Remove line returns here to keep lines short.
  sed -Ez 's/\\\n//Mg' "./${LFBFL_subdir2}/files_names_listing.txt"\
    | grep -v '^//'\
    | while read -r LFBFL_file_name;
  do
    echo "Listing file for tex/HTML : ${LFBFL_file_name}"
    # echo "Listing file for tex : ${LFBFL_file_name}"
    # LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    LFBFL_cleaned_path1="${LFBFL_file_name//_/\\_}"
    LFBFL_cleaned_path2=$(
      echo "${LFBFL_file_name}"\
      | sed -e 's/\//:/g' -e 's/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"\
        "\\"
```

```
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        "${LFBFL_score_command_properties2}"
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_name}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    {
      echo "\subsection{"
      echo "  ${LFBFL_file_name}"\
        | sed -e "s|  ${LFBFL_file_name}|${LFBFL_new_lines}|g"\
              -e 's/_/\\_/g'
      echo "}"
      echo "\label{"
      echo "  ${LFBFL_cleaned_path2}"\
        | sed -e "s|  ${LFBFL_cleaned_path2}|${LFBFL_new_lines2}|g"
      echo "}"
      echo ""
      echo "\VerbatimInput[numbers=left,xleftmargin=-5mm]{"
      echo "  ${LFBFL_file_name}"\
        | sed -e "s|  ${LFBFL_file_name}|${LFBFL_new_lines3}|g"
      echo "}"
      echo ""
      echo ""
    } >> "${LFBFL_temp_files_listing}"

    # echo "Listing file for HTML : ${LFBFL_file_name}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_name}"
```

```
    if [[ ${#LFBFL_file_name} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      echo "<h3 id=\"subsection2.${LFBFL_i}\">2.${LFBFL_i}"
      # shellcheck disable=SC2001
      echo "${LFBFL_file_name}"\
        | sed -e "s|${LFBFL_file_name}|${LFBFL_new_lines}|g"
      echo "</h3>"
      echo "<pre class=\"numbered_lines\">"
      sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g'\
        < "${LFBFL_file_name}"
      echo "</pre>"
      echo ""
      echo ""
    } >> "${LFBFL_temp_files_listing2}"
    {
      echo "      <li><a href=\"#subsection2.${LFBFL_i}\">"
      # shellcheck disable=SC2001
      echo "${LFBFL_file_name}"\
        | sed -e "s|${LFBFL_file_name}|${LFBFL_new_lines}|g"
      echo "      </a></li>"
    } >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "./${LFBFL_subdir2}/files_listing.tex.tpl"\
    "${LFBFL_temp_files_listing}"
  sed -i -e "s|\\n</pre>|</pre>|g" "${LFBFL_temp_files_listing2}"
  overwrite_if_not_equal\
    "./${LFBFL_subdir2}/temp/files_listing.html.tpl"\
    "${LFBFL_temp_files_listing2}"

  # We verify if some lines are beyond max_line_length characters
  # in current_tree_light.txt and current_tree.txt.
  declare -ar LFBFL_trees=(
    "current_tree_light.txt"
    "current_tree.txt.temp"
  )
  local LFBFL_tree_path
  local LFBFL_line
```

```
  local LFBFL_prefix
  declare -i LFBFL_prefix_last_position
  local LFBFL_char
  local LFBFL_line_start
  declare -ir LFBFL_overlength=$((LFBFL_max_line_length+1))
  local LFBFL_file_name2
  for LFBFL_tree in "${LFBFL_trees[@]}"; do
    LFBFL_tree_path="${LFBFL_subdir2}/temp/${LFBFL_tree}"
    # shellcheck disable=SC2248
    grep '.\{'${LFBFL_overlength}'\}' "${LFBFL_tree_path}"\
      | while read -r LFBFL_line;
    do
      # echo "LFBFL_line: ${LFBFL_line}"
      LFBFL_prefix=$(
        echo "${LFBFL_line}"\
        | sed -E -e 's/(.*)──[^─]+$/\1/g'
      )
      # echo "LFBFL_prefix: ${LFBFL_prefix}"
      LFBFL_prefix_last_position=$((${#LFBFL_prefix}-1))
      LFBFL_char="${LFBFL_prefix:${LFBFL_prefix_last_position}:1}"
      # echo "LFBFL_char: ${LFBFL_char}"
      LFBFL_prefix=$(
        echo "${LFBFL_prefix}"\
        | sed -E -e 's/[^ ]+$//g'
      )
      if [[ "${LFBFL_char}" == "└" ]]; then
        LFBFL_prefix+="  "
      else
        LFBFL_prefix+="│ "
      fi
      # echo "LFBFL_prefix: ${LFBFL_prefix}"
      LFBFL_file_name=$(
        echo "${LFBFL_line}"\
        | sed -E 's|.* (([a-zA-Z0-9\._/-]+).)$|\1|g'
      )
      # echo "LFBFL_file_name: ${LFBFL_file_name}"
      LFBFL_file_name2=$(
        echo "${LFBFL_file_name}"\
        | sed -E -e 's/\*/\\\*/g'
      )
      LFBFL_line_start=$(
        echo "${LFBFL_line}"\
        | sed -E -e "s/(.*)[ ]*${LFBFL_file_name2}/\1/g"\
                 -e 's/ *$//g'
      )
      # echo "LFBFL_line_start: ${LFBFL_line_start}"
```

```
      LFBFL_line=$(
        echo "${LFBFL_line}"\
        | sed -E -e 's/\[/\\\[/g' -e 's/\]/\\\]/g' -e 's/\*/\\\*/g'
      )
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      sed -i -e\
        "s/${LFBFL_line}/${LFBFL_line_start}\n${LFBFL_new_lines}/g"\
        "${LFBFL_tree_path}"
    done
  done

  overwrite_if_not_equal "${LFBFL_subdir2}/temp/current_tree.txt"\
    "${LFBFL_subdir2}/temp/current_tree.txt.temp" 1 1

  declare -r LFBFL_tex_path_start=\
"./${LFBFL_subdir2}/temp/${LFBFL_repository_name}.tex"
  declare -r LFBFL_html_path_start=\
"./${LFBFL_subdir2}/temp/${LFBFL_repository_name}.html"

  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]];
  then
    # If there is a template, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]];
  then
    # Otherwise if there is a tex file, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
    echo "Neither .tex.tpl, nor .tex in ./${LFBFL_subdir2}/"
  fi

  # Same logic with repository HTML file.
  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]];
  then
```

```
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
  elif [[ -f "./${LFBFL_repository_name}.html" ]]; then
    cp "./${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    echo "Neither .html.tpl in ./${LFBFL_subdir2}/, nor .html in ./"
  fi

  # shellcheck disable=SC2001
  echo "${LFBFL_abstract}"\
    | sed -e 's/\\n/\n/g'\
    > "./${LFBFL_subdir2}/temp/abstract_temp"

  # shellcheck disable=SC2001
  echo "${LFBFL_acknowledgments}"\
    | sed -e 's/\\n/\n/g'\
    > "./${LFBFL_subdir2}/temp/acknowledgments_temp"

  local LFBFL_temp_path
  LFBFL_temp_path=$(realpath "./${LFBFL_subdir2}/temp/")
  readonly LFBFL_temp_path

  # But the filling still occurs, in case the dev want to refill
  # part of the tex/html file that he modified with @token@
  # using some computed results.
  # Tex filling:
  if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
           -e "s|@number_of_commits@|${LFBFL_number_of_commits}|g"\
           -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_tex_path_start}.1"

    insert_file_at_token "${LFBFL_tex_path_start}.1" @abstract@\
      "./${LFBFL_subdir2}/temp/abstract_temp"\
      "${LFBFL_tex_path_start}.2"

    insert_file_at_token "${LFBFL_tex_path_start}.2"\
      @acknowledgments@\
      "./${LFBFL_subdir2}/temp/acknowledgments_temp"\
```

```
      "${LFBFL_tex_path_start}.3"

    enhanced_pushd "${LFBFL_temp_path}" 2
    can_continue_after_enhanced_pushd || return
    sed -i\
      -e '/@current_tree_light@/{r current_tree_light.txt' -e 'd}'\
      -e '/@current_tree@/{r current_tree.txt' -e 'd}'\
      "${LFBFL_repository_name}.tex.3"
    if [[ enhanced_pushd_result -eq 0 ]]; then
      enhanced_popd "${LFBFL_temp_path}" 2\
        || return
    fi

    insert_file_at_token "${LFBFL_tex_path_start}.3"\
      @files_listing_VerbatimInput@\
      "./${LFBFL_subdir2}/files_listing.tex.tpl"\
      "${LFBFL_tex_path_start}.4"

    overwrite_if_not_equal\
      "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.4" 1
  fi

  # HTML filling:
  declare -i LFBFL_HTML_updated=0
  if [[ -f "${LFBFL_html_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
           -e "s|@number_of_commits@|${LFBFL_number_of_commits}|g"\
           -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_html_path_start}.1"

    insert_file_at_token "${LFBFL_html_path_start}.1" @abstract@\
      "./${LFBFL_subdir2}/temp/abstract_temp"\
      "${LFBFL_html_path_start}.2"

    insert_file_at_token "${LFBFL_html_path_start}.2"\
      @acknowledgments@\
      "./${LFBFL_subdir2}/temp/acknowledgments_temp"\
      "${LFBFL_html_path_start}.3"
```

```
    insert_file_at_token "${LFBFL_html_path_start}.3"\
      @files_lis@ "${LFBFL_temp_files_lis}"\
      "${LFBFL_html_path_start}.4"

    enhanced_pushd "${LFBFL_temp_path}" 2
    can_continue_after_enhanced_pushd || return
    sed -i\
      -e '/@current_tree_light@/{r current_tree_light.txt' -e 'd}'\
      -e '/@current_tree@/{r current_tree.txt' -e 'd}'\
      "${LFBFL_repository_name}.html.4"
    if [[ enhanced_pushd_result -eq 0 ]]; then
      enhanced_popd "${LFBFL_temp_path}" 2\
        || return
    fi

    insert_file_at_token "${LFBFL_html_path_start}.4"\
      @files_listing_HTMLPreInput@\
      "./${LFBFL_subdir2}/temp/files_listing.html.tpl"\
      "${LFBFL_html_path_start}.5"

    overwrite_if_not_equal "./${LFBFL_repository_name}.html"\
      "${LFBFL_html_path_start}.5" 1
    LFBFL_HTML_updated=$?
  fi

  # The HTML contains everything including listings with source code.
  # If it doesn't get updated,
  # then it is useless to recompute the PDF.
  if [[ LFBFL_HTML_updated -eq 0 ]]; then
    return
  fi

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    for ((i=0; i<3; i++)); do
      pdflatex "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"
    done
  else
    for ((i=0; i<3; i++)); do
      pdflatex "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
        > /dev/null
    done
  fi

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
```

```
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv "${LFBFL_file_name}" "${LFBFL_subdir2}/temp/"
  done
}

create_PDF "$@"
```

2.11 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
```

```
  # $1=base_file_name
  # $2=target_file_name
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    echo "$3"
    cat "$1"
    echo "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
  # $1=base_file_name
  # $2=target_file_name
  # $3=line_comment_prefix
  # $4=optional_post_processing
  # $5=file_prefix
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
  readonly LFBFL_temp
  if [[ -n "$5" ]]; then
    echo "$5"\
      > "${LFBFL_temp}"
  fi
  sed -e "s/^/$3/g" "$1"\
    >> "${LFBFL_temp}"
  if [[ -n "$4" ]]; then
    eval "$4"
  fi
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

split_file_in_two(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_part1
  # $4=$file_name_part2
  declare -ir LFBFL_line_number=$(
    grep -n "$2" "$1"\
    | head --lines=1\
    | cut -f 1 -d ':'
  )
```

```
  declare -ir LFBFL_line_count=$(ll_wc -l -n "$1")
  declare -ir LFBFL_lines_after=$((
    LFBFL_line_count - LFBFL_line_number
  ))
  head --lines="$((LFBFL_line_number - 1))" "$1"\
    > "$3"
  tail --lines="${LFBFL_lines_after}" "$1"\
    > "$4"
}

insert_file_at_token(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_to_insert
  # $4=$file_name_result
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
    "${LFBFL_end_file_name}"
  cat "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}"
  fi
}
```

2.12 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
```

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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
```

```
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

2.13 build_and_checks_dependencies/grammar_and_spell_check.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spell_check(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && echo "pipefail option activated"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
```

```
  local LFBFL_grammar_or_spell_checker_command=""
  grep_variable "$1" grammar_or_spell_checker_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_or_spell_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    return
  fi
  local LFBFL_file_path
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 0 ]]\
      || echo "Iterating on pattern: ${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | while read -r LFBFL_file_path;
    do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval "${LFBFL_eval_string}"
    done
  done
}
```

2.14 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
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```

```
#
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# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && echo "pipefail option activated"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  # Source
  local LFBFL_license_source_subdir
  LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
  readonly LFBFL_license_source_subdir
  local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
  LFBFL_source_license_prefix+="license_file_header_"
  readonly LFBFL_source_license_prefix
  # Target
  local LFBFL_license_target_subdir
  LFBFL_license_target_subdir="${LFBFL_work_directory}/"
  LFBFL_license_target_subdir="build_and_checks_variables/temp/"
  readonly LFBFL_license_target_subdir
  local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
  LFBFL_target_license_prefix+="license_file_header_"
```

```
readonly LFBFL_target_license_prefix

declare -ar LFBFL_licenses=(
  "AGPLv3+"
  "GPLv3+"
  "LGPLv3+"
)

# ------------------------------------------------------------------
# Various language definitions for text file formats
declare -ar LFBFL_block_comment_languages=(
  "c"
  "py"
)
declare -ar LFBFL_block_comment_enters=(
  '/*'
  '"""'
)
declare -ar LFBFL_block_comment_exits=(
  '*/'
  '"""'
)
equal "${#LFBFL_block_comment_languages[@]}"\
      "${#LFBFL_block_comment_enters[@]}"\
      "${#LFBFL_block_comment_exits[@]}"
declare -i LFBFL_equal_result=$?
if [[ LFBFL_equal_result -eq 0 ]]; then
  echo $'/!\\Problem with the definitions of languages arrays 1/!\\'
fi

declare -ar LFBFL_line_comment_languages=(
  "sh"
  "tex"
)
declare -ar LFBFL_line_comment_prefixes=(
  '# '
  '% '
)
equal "${#LFBFL_line_comment_languages[@]}"\
      "${#LFBFL_line_comment_prefixes[@]}"
LFBFL_equal_result=$?
if [[ LFBFL_equal_result -eq 0 ]]; then
  echo $'/!\\Problem with the definitions of languages arrays 2/!\\'
fi
# ------------------------------------------------------------------

```

```
114     # ------------------------------------------------------------------
115     # We generate the license header templates adapted to various text
116     # files formats.
117     # Kamoulox do endfor, Jacques Beauregard XD
118     # (C'est toujours mieux que l'inverse :) !)
119     # for do endfor -> for do done (Fort Doux Donne).
120     local LFBFL_some_license
121     local LFBFL_license_prefix
122     local LFBFL_license_prefix2
123     local LFBFL_extension
124     local LFBFL_license_file_name
125     local LFBFL_enter_string
126     local LFBFL_exit_string
127     local LFBFL_temp2
128     local LFBFL_prefix_string
129     local LFBFL_intermediate_file_name
130     local LFBFL_file_prefix
131     declare -i LFBFL_generate_from_template_result
132     for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
133       LFBFL_license_prefix="${LFBFL_source_license_prefix}"
134       LFBFL_license_prefix+="${LFBFL_some_license}"
135       LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
136       LFBFL_license_prefix2+="${LFBFL_some_license}"
137       for ((i=0; i<${#LFBFL_block_comment_languages[@]}; i++)); do
138         LFBFL_extension=${LFBFL_block_comment_languages[i]}
139         LFBFL_license_file_name="${LFBFL_license_prefix2}"
140         LFBFL_license_file_name+=".${LFBFL_extension}"
141         LFBFL_enter_string=${LFBFL_block_comment_enters[i]}
142         LFBFL_exit_string=${LFBFL_block_comment_exits[i]}
143         generate_from_template_with_block_comments\
144           "${LFBFL_license_prefix}.tpl"\
145           "${LFBFL_license_file_name}.tpl"\
146           "${LFBFL_enter_string}" "${LFBFL_exit_string}"
147         LFBFL_generate_from_template_result=$?
148         if [[ LFBFL_i_verbose -eq 1 ]]; then
149           if [[ LFBFL_generate_from_template_result -eq 1 ]]; then
150             echo\
151               "License template ${LFBFL_license_file_name}.tpl updated"
152           elif [[ LFBFL_generate_from_template_result -eq 2 ]]; then
153             echo\
154               "License template ${LFBFL_license_file_name}.tpl created"
155           fi
156         fi
157       done
158       LFBFL_temp2=".generate_from_template_with_line_comments.temp"
159       for ((i=0; i<${#LFBFL_line_comment_languages[@]}; i++)); do
```

```
      LFBFL_extension=${LFBFL_line_comment_languages[i]}
      LFBFL_license_file_name="${LFBFL_license_prefix2}"
      LFBFL_license_file_name+=".${LFBFL_extension}"
      LFBFL_prefix_string=${LFBFL_line_comment_prefixes[i]}
      LFBFL_intermediate_file_name="${LFBFL_license_file_name}.tpl"
      LFBFL_intermediate_file_name+="${LFBFL_temp2}"
      LFBFL_file_prefix=""
      if [[ "${LFBFL_extension}" == "sh" ]]; then
        LFBFL_file_prefix="#!/usr/bin/env bash"
      fi
      generate_from_template_with_line_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_name}.tpl"\
        "${LFBFL_prefix_string}"\
        "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_name}'"\
        "${LFBFL_file_prefix}"
      LFBFL_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_generate_from_template_result -eq 1 ]]; then
          echo\
            "License template ${LFBFL_license_file_name}.tpl updated"
        elif [[ LFBFL_generate_from_template_result -eq 2 ]]; then
          echo\
            "License template ${LFBFL_license_file_name}.tpl created"
        fi
      fi
    done
  done
  # ------------------------------------------------------------------

  # Now that base license templates generated languages license
  # templates in working directory, we can go there.
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  local LFBFL_data_file_name="build_and_checks_variables/"
  LFBFL_data_file_name+="repository_data.txt"
  readonly LFBFL_data_file_name
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  local LFBFL_license=""
  grep_variable "${LFBFL_data_file_name}" license\
    --result-variable-prefix="LFBFL_"
```

```
206   local LFBFL_license2=""
207   grep_variable "${LFBFL_data_file_name}" license2\
208     --result-variable-prefix="LFBFL_"
209   local LFBFL_author_full_name=""
210   grep_variable "${LFBFL_data_file_name}" author_full_name\
211     --result-variable-prefix="LFBFL_"\
212     --replace-line-returns-by=" "
213
214   LFBFL_license_prefix="build_and_checks_variables/temp/"
215   LFBFL_license_prefix+="license_file_header_"
216   readonly LFBFL_license_prefix
217   LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
218   readonly LFBFL_license_prefix2
219   declare -r\
220     LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
221
222   # First year according to current state of git repository.
223   declare -ir LFBFL_first_year=$(
224     git log\
225     | grep 'Date:'\
226     | cut -f 8 -d ' '\
227     | tail -1
228   )
229   # Last year according to current state of git repository.
230   declare -ir LFBFL_last_year=$(
231     git log\
232     | grep 'Date:'\
233     | cut -f 8 -d ' '\
234     | head -1
235   )
236   local LFBFL_copyright_string
237   LFBFL_copyright_string="${LFBFL_first_year}-${LFBFL_last_year}"
238   LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
239   readonly LFBFL_copyright_string
240
241   prepare_filled_license_file_block(){
242     # $1=license_file_name
243     sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
244         -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
245         "$1.tpl"\
246       > "$1.temp"
247     overwrite_if_not_equal "$1" "$1.temp"
248     declare -r LFBFL_result=$?
249     if [[ LFBFL_i_verbose -eq 1 ]]; then
250       if [[ LFBFL_result -eq 1 ]]; then
251         echo "License template $1 updated"
```

```
      elif [[ LFBFL_result -eq 2 ]]; then
        echo "License template $1 created"
      fi
    fi
    head --lines=-1 "$1"\
      | tail --lines=+2\
      > "$1.temp"
    # Strange world: tail --lines=-1 was working
    # to exclude first line.
    # I tested it months ago, and saw with my eyes the start and end
    # block comments removed with:
    # head --lines=-1 "$1" | tail --lines=-1.
    # Now, man page says to use --lines=+2 instead.
    # Welcome in the world where AI change your environment and your
    # documentation on the fly...
  }

  declare -Ar LFBFL_all_block_comment_languages=(
    ["c"]="c"
    ["php"]="c"
    ["py"]="py"
  )

  local LFBFL_key
  local LFBFL_dest
  local LFBFL_license_file_name2
  declare -i LFBFL_not_subfile
  declare -i LFBFL_not_subfile2
  for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_block "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_block "${LFBFL_license_file_name2}"
    fi
    find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
      | relevant_find\
      | while read -r LFBFL_file_name;
    do
      is_subfile "${LFBFL_file_name}"\
        "${LFBFL_license_file_name}.temp"
      LFBFL_not_subfile=$?
      LFBFL_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
```

```
298         is_subfile "${LFBFL_file_name}"\
299           "${LFBFL_license_file_name2}.temp"
300         LFBFL_not_subfile2=$?
301       fi
302       if [[ LFBFL_not_subfile -ge 1 && LFBFL_not_subfile2 -ge 1 ]];
303       then
304         echo "File ${LFBFL_file_name} has no/wrong license header?"
305       fi
306     done
307     rm "${LFBFL_license_file_name}.temp"
308     if [[ -n "${LFBFL_license2}" ]]; then
309       rm "${LFBFL_license_file_name2}.temp"
310     fi
311   done
312
313   prepare_filled_license_file_line(){
314     # $1=LFBFL_license_file_name
315     sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
316         -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
317         "$1.tpl"\
318       > "$1.temp"
319     overwrite_if_not_equal "$1" "$1.temp"
320     declare -r LFBFL_result=$?
321     if [[ LFBFL_i_verbose -eq 1 ]]; then
322       if [[ LFBFL_result -eq 1 ]]; then
323         echo "License template $1 updated"
324       elif [[ LFBFL_result -eq 2 ]]; then
325         echo "License template $1 created"
326       fi
327     fi
328   }
329
330   declare -Ar LFBFL_all_line_comment_languages=(
331     ["sh"]="sh"
332     ["tex"]="tex"
333   )
334
335   for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
336     LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
337     LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
338     prepare_filled_license_file_line "${LFBFL_license_file_name}"
339     if [[ -n "${LFBFL_license2}" ]]; then
340       LFBFL_license_file_name2="${LFBFL_license_prefix3}"
341       LFBFL_license_file_name2+=".${LFBFL_dest}"
342       prepare_filled_license_file_line "${LFBFL_license_file_name2}"
343     fi
```

```
    find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
      | relevant_find\
      | while read -r LFBFL_file_name;
    do
      is_subfile "${LFBFL_file_name}" "${LFBFL_license_file_name}"
      LFBFL_not_subfile=$?
      LFBFL_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_name}" "${LFBFL_license_file_name2}"
        LFBFL_not_subfile2=$?
      fi
      if [[ LFBFL_not_subfile -ge 1 && LFBFL_not_subfile2 -ge 1 ]];
      then
        echo "File ${LFBFL_file_name} has no/wrong license header?"
      fi
    done
  done
}

build_licenses_templates "$@"
```

2.15 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```
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```

2.16 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

2.17 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

```
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2.18 build_and_checks_dependencies/lines_counts.libr.sh

```
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#
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# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
```

```
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && echo "Iterating on pattern: ${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H -v 'a(?!a)a'
  done
}

all_self_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN\
    || return

  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
  # below and another lines, see SC2145 false positive
  all_code_lines "$@" \
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

```

```
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
  all_self_code_lines "$@" \
    | empty_lines_after_file_name
}

all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
  all_self_code_lines "$@" \
    | not_empty_lines_after_file_name
}

code_lines_count_all(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
  all_self_code_lines "$@" \
    | wc -l
}

code_lines_count_empty(){
```

```
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
  all_self_empty_code_lines "$@" \
    | wc -l
}

code_lines_count_not_empty(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
  all_self_not_empty_code_lines "$@" \
    | wc -l
}
```

2.19 build_and_checks_dependencies/lines_filters.libr.sh

```
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# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # echo "$ll_wc_var_arg"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_number_only=1
```

```
    elif [[ ll_wc_var_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # echo "$ll_wc_var_number_only"
  if [[ ll_wc_var_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # echo "normal"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# echo "https://domain/list-path/2024-05/msg00013.html" | \
#   sed -e 's|domain|lists.gnu.org|' \
#       -e 's|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
```

```
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
  declare -ir LFBFL_lines_before=$(ll_wc -l -n "${!#}")
  declare -ir LFBFL_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # echo "$LFBFL_lines_before"
  # echo "$LFBFL_lines_after"
  if [[ "${LFBFL_lines_before}" ==  "${LFBFL_lines_after}" ]]; then
    rm "${LFBFL_temp}"
    return
  fi
  mv "${LFBFL_temp}" "${!#}"
}

grep_variable(){
  # $1=$file
  # $2=$variable_name
  # Options :
  #   --replace-line-returns-by=""
  #   --result-variable-prefix="LFBFL_" for example
  declare -r LFBFL_regexp="(?<=^$2=).*$"
  local LFBFL_arg
  # echo $LFBFL_regexp
  local LFBFL_variable_value
  if [[ "$*" == *--replace-line-returns-by=* ]]; then
    local LFBFL_s_replace=""
    for LFBFL_arg in "$@"; do
      if [[ "${LFBFL_arg}" == --replace-line-returns-by=* ]]; then
        LFBFL_s_replace=${LFBFL_arg#--replace-line-returns-by=}
        break
      fi
    done
    LFBFL_variable_value=$(
      grep -oP "${LFBFL_regexp}" "$1"\
      | sed -Ez -e 's/\n/'"${LFBFL_s_replace}"'/Mg'
    )
    # We remove the trailing replace string,
```

```
154     # since the intent is only to replace the intermediate \n.
155     if [[ -n "${LFBFL_s_replace}" ]]; then
156     LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
157     fi
158   else
159     LFBFL_variable_value=$(grep -oP "${LFBFL_regexp}" "$1")
160   fi
161   local LFBFL_prefix=""
162   for LFBFL_arg in "$@"; do
163     if [[ "${LFBFL_arg}" == --result-variable-prefix=* ]]; then
164       LFBFL_prefix=${LFBFL_arg#--result-variable-prefix=}
165       break
166     fi
167   done
168   declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
169   # echo "${LFBFL_variable_name} : ${LFBFL_variable_value}"
170   # The declare line may be dropped by a local or declare
171   # in outer scope.
172   # declare -g "${LFBFL_variable_name}"
173   # So the assignment must be done on another line.
174   printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
175   # echo "${!LFBFL_variable_name}"
176 }
177
178 empty_lines(){
179   grep '^$'
180 }
181
182 not_empty_lines(){
183   grep -v '^$'
184 }
185
186 space_starting_lines(){
187   grep '^[ ]'
188 }
189
190 not_space_starting_lines(){
191   grep '^[^ ]'
192 }
193
194 empty_lines_after_file_name(){
195   grep '^[^:]\+:$'
196 }
197
198 not_empty_lines_after_file_name(){
199   grep -v '^[^:]\+:$'
```

```
}

space_starting_lines_after_file_name(){
  grep '^[^:]\+: '
}

not_space_starting_lines_after_file_name(){
  grep '^[^:]\+:[^ ]'
}

not_JS_dependencies_find(){
  grep -vE -e "(^|/)node_modules/"\
           -e "(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep -vE -e "^([^:]+/)?node_modules/"\
           -e "^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
  not_JS_dependencies_grep
}

not_python_cache_find(){
  grep -vE -e "(^|/)__pycache__/"\
           -e "(^|/)\.mypy_cache/"\
           -e "(^|/)\.ruff_cache/"
}

not_python_cache_grep(){
  grep -vE -e "^([^:]+/)?__pycache__/"\
           -e "^([^:]+/)?\.mypy_cache/"\
           -e "^([^:]+/)?\.ruff_cache/"
}

not_cache_find(){
  not_python_cache_find
}

not_cache_grep(){
  not_python_cache_grep
```

```
}

not_git_find(){
  grep -vE "(^|/)\.git/"
}

not_git_grep(){
  grep -vE "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep -vE "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep -vE "^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep -vE -e "(^|/)COPYING$"\
           -e "(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep -vE -e "^([^:]+/)?COPYING:"\
           -e "^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

```

```
292 not_main_html_find(){
293   local LFBFL_repository_name=""
294   grep_variable build_and_checks_variables/repository_data.txt\
295     repository_name\
296     --result-variable-prefix="LFBFL_"\
297     --replace-line-returns-by=""
298   grep -vE "(^|/)${LFBFL_repository_name}\.html$"
299 }
300
301 not_main_html_grep(){
302   local LFBFL_repository_name=""
303   grep_variable build_and_checks_variables/repository_data.txt\
304     repository_name\
305     --result-variable-prefix="LFBFL_"\
306     --replace-line-returns-by=""
307   grep -vE "^([^:]+/)?${LFBFL_repository_name}\.html:"
308 }
309
310 not_temp_file_find(){
311   grep -vE "(^|/)build_and_checks_variables/temp/.*$"
312 }
313
314 not_temp_file_grep(){
315   grep -vE "^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
316 }
317
318 relevant_find(){
319   if [[ ! -o pipefail ]]; then
320     set -o pipefail
321     trap 'set +o pipefail' RETURN
322   fi
323   not_dependencies_find\
324     | not_cache_find\
325     | not_git_find\
326     | not_archive_find\
327     | not_temp_file_find
328 }
329
330 relevant_grep(){
331   if [[ ! -o pipefail ]]; then
332     set -o pipefail
333     trap 'set +o pipefail' RETURN
334   fi
335   not_dependencies_grep\
336     | not_cache_grep\
337     | not_git_grep\
```

```
    | not_archive_grep\
    | not_temp_file_grep
}
```

2.20 build_and_checks_dependencies/lines_maps.libr.sh

```
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# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed -e 's/\xc2\xa0/ /g'\
      -e 's/\xe2\x80\x80/ /g'\
      -e 's/\xe2\x80\x81/ /g'\
```

```
        -e 's/\xe2\x80\x82/ /g'\
        -e 's/\xe2\x80\x83/ /g'\
        -e 's/\xe2\x80\x84/ /g'\
        -e 's/\xe2\x80\x85/ /g'\
        -e 's/\xe2\x80\x86/ /g'\
        -e 's/\xe2\x80\x87/ /g'\
        -e 's/\xe2\x80\x88/ /g'\
        -e 's/\xe2\x80\x89/ /g'\
        -e 's/\xe2\x80\x8a/ /g'\
        -e 's/\xe2\x80\x8b/ /g'\
        -e 's/\xe2\x80\x8c/ /g'\
        -e 's/\xe2\x80\x8d/ /g'\
        -e 's/\xe2\x80\x8e/ /g'\
        -e 's/\xe2\x80\x8f/ /g'\
        -e 's/\xe2\x80\xa8/ /g'\
        -e 's/\xe2\x80\xa9/ /g'\
        -e 's/\xe2\x80\xaa/ /g'\
        -e 's/\xe2\x80\xab/ /g'\
        -e 's/\xe2\x80\xac/ /g'\
        -e 's/\xe2\x80\xad/ /g'\
        -e 's/\xe2\x80\xae/ /g'\
        -e 's/\xe2\x80\xaf/ /g'\
        -e 's/\xe2\x81\x9f/ /g'\
        -e 's/\xe2\x81\xa0/ /g'\
        -e 's/\xe2\x81\xa1/ /g'\
        -e 's/\xe2\x81\xa2/ /g'\
        -e 's/\xe2\x81\xa3/ /g'\
        -e 's/\xe2\x81\xa4/ /g'\
        -e 's/\xe2\x81\xa5/ /g'\
        -e 's/\xe2\x81\xa6/ /g'\
        -e 's/\xe2\x81\xa7/ /g'\
        -e 's/\xe2\x81\xa8/ /g'\
        -e 's/\xe2\x81\xa9/ /g'\
        -e 's/\xe2\x81\xaa/ /g'\
        -e 's/\xe2\x81\xab/ /g'\
        -e 's/\xe2\x81\xac/ /g'\
        -e 's/\xe2\x81\xad/ /g'\
        -e 's/\xe2\x81\xae/ /g'\
        -e 's/\xe2\x81\xaf/ /g'
}
```

2.21 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
  # $1=$target_file_name
  # $2=$temp_file_name
  # $3=$keep_temp
  # $4=$tree_mode : when the two files are the output of tree command,
  # we ignore if all the differences are only on directories.
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  if [[ ! -f "$1" ]]; then
    if [[ -n "$3" ]]; then
      cp -p "$2" "$1"
    else
      mv "$2" "$1"
    fi
    return 2
  fi
  declare -i LFBFL_is_equal
  if [[ -n "$4" ]]; then
    # Not setting pipefail since the result would still be incorrect.
    # shellcheck disable=SC2312
    diff "$1" "$2"\
      | grep -E "^(>|<)"\
```

```
      | grep -v "/$"
    LFBFL_is_equal=${PIPESTATUS[2]}
  else
    diff -q "$1" "$2"
    LFBFL_is_equal=1-$?
  fi
  if [[ LFBFL_is_equal -eq 1 ]]; then
    if [[ -z "$3" ]]; then
      rm "$2"
    fi
    return 0
  fi
  if [[ -n "$3" ]]; then
    cp -p "$2" "$1"
  else
    mv "$2" "$1"
  fi
  return 1
}
```

2.22 build_and_checks_dependencies/python_black_complement.libr.sh

```
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#
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# This file was renamed from "python_black_complement.sh"
```

```
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  [[ LFBFL_i_verbose -eq 1 ]]\
    && echo "Checking empty lines after Python function docstrings"
  pcre2grep -M --\
    $'def ([^"]|"(?!"))*"""([^"]|"(?!"))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  check_no_empty_line_after_python_function_docstring "$@"
}
```

2.23 build_and_checks_dependencies/python_isort_complement.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
```

```
  can_continue_after_enhanced_pushd || return

  [[ LFBFL_i_verbose -eq 1 ]]\
    && echo "Checking import of _collections_abc is at the right place"
  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
  declare -r LFBFL_temp=".check_collections_abc_place.temp"
  local LFBFL_file_name
  find . -name "*.py"\
    | relevant_find\
    | while read -r LFBFL_file_name;
  do
    [[ -f "${LFBFL_file_name}" ]] || continue
    sed -Ez 's/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
      "${LFBFL_file_name}"\
      > "${LFBFL_file_name}${LFBFL_temp}"
    overwrite_if_not_equal "${LFBFL_file_name}"\
      "${LFBFL_file_name}${LFBFL_temp}"
  done
}

python_isort_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  check_collections_abc_place "$@"
}
```

2.24 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php
/*
This file is part of DevOrSysAdminScripts library.

```

```
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".
*/

require_once('./build_and_checks_dependencies/split_score.libr.php');

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// The first cast "(int) generate_split_score" is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
```

```
    (bool) $argv[1], (int) $argv[2], explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
```

2.25 build_and_checks_dependencies/split_score.libr.php

```
<?php
/*
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".
*/

function generate_split_score(
  $b_after_before,
  $i_max_length,
  $a_delimiter_strings_domain,
){
  return function(
    $s_delimiter_string,
    $i_cut_position,
```

```
    $b_is_cut_after,
  ) use ($b_after_before, $i_max_length, $a_delimiter_strings_domain){
    if(
      !in_array(
        $s_delimiter_string, $a_delimiter_strings_domain, true
      )
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_after_before){
      // When $b_after_before,
      // always way better to cut after '/' than before it.
      // When !$b_after_before (aka before_after),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
}
```

2.26 build_and_checks_dependencies/strings_functions.libr.sh

```
#!/usr/bin/env bash
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```

```
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#
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# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"

split_line_at(){
  # $1=$line
  # $2=$position
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # echo "$split_line_at_result_beginning"
  # echo "$split_line_at_result_end"
}

split_score_simple(){
  # $1=$after_before
  # $2=$delimiters_strings_domain concatenated characters/delimiters
  # $3=$delimiter_string a single character
  # $4=$is_cut_after
  declare -gi split_score_result
  if [[ "$2" != *$3* ]]; then
    split_score_result=0
    return
  fi
  if [[ "$4" == "$1" ]]; then
    split_score_result=2
    return
  fi
  split_score_result=1
}

get_split_score_simple(){
  # $1=$after_before
  # $2=$max_length
  # $3=$delimiters_strings_domain
```

```
  declare -g get_split_score_result="split_score_simple $1 '$3'"
  declare -gi get_split_score_result2
  get_split_score_result2=$((7+$1*4))
  # if [[ $1 -eq 1 ]]; then
  #   get_split_score_result2=7
  # else
  #   get_split_score_result2=11
  # fi
}

get_split_score_exec(){
  declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
  SPLIT_SCORE_EXEC+="/split_score.exec.php"
}

split_score(){
  # $1=$after_before
  # $2=$max_length
  # $3=$delimiters_strings_domain concatenated characters/delimiters
  # $4=$delimiter_string a single character
  # $5=$cut_position
  # $6=$is_cut_after
  get_split_score_exec
  declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
  declare -gi split_score_result
  split_score_result=$(eval "${LFBFL_command}")
}

# This function is currently not called,
# hence we need that shellcheck disable.
# shellcheck disable=SC2034
get_split_score(){
  # $1=$after_before
  # $1=$max_length
  # $2=$delimiters_strings_domain
  declare -g get_split_score_result="split_score $1 $2 '$3'"
  declare -gi get_split_score_result2
  get_split_score_result2=$((5+$1*4))
}

split_line_at_most(){
  # $1=$line
  # $2=$max_length of beginning result string
  # $3=$split_score_command
  #   negative score means do not consider as a possible split
  # $4=$split_score_command_properties :
```

```
  #   These properties are flags.
  #   If someday some property is not a flag,
  #   then create split_score_command_properties2.
  #   - null or 0 no property;
  #   - 1 split score doesn't depend on which delimiter matches;
  #   - 2 split score doesn't depend on the current position;
  #   - 4 any split score after is always larger
  #     than any split score before;
  #   - 8 any split score before is always larger
  #     than any split score after;
  # $5=$forbidden_previous_character
  # $6=$split_score_command_delimiter_strings_domain
  #   optimisation if small domain
  #   - this argument can be "null", then we will use $3 only instead,
  #     in that case, only mono-character strings are considered,
  #   - this argument can be an array of characters,
  #     optimize accordingly,
  #   - this argument can be an array of strings,
  #     optimize accordingly,
  #   - this argument can be an array of regexps,
  #     optimize accordingly.
  #     Not sure an array of regexps makes more sense than a single
  #     one. Maybe they could be both cases with corresponding
  #     optimizations. Same for a single character or substring.
  #
  # Note that we're always looking for the last position to split
  # among maximum score;
  # if some position gets as score from 2 delimiters,
  # its score is always the maximum of the two corresponding scores.
  declare -g split_line_at_most_result_start
  declare -g split_line_at_most_result_end
  declare -A LFBFL_positions
  # echo "$1 $2 $3 $4"
  # For my use case in bash scripts, I will need only an array of
  # characters. See get_split_score_after_before().
  if [[ -n "$6" ]]; then
    echo "split_line_at_most() \$4 NOT IMPLEMENTED YET"
  fi
  if [[ $2 -ge ${#1} ]]; then
    split_line_at_most_result_start=$1
    split_line_at_most_result_end=""
    return
  fi
  declare -r LFBFL_sort_command="sort --numeric-sort"
  declare -ir LFBFL_length_minus_1=$(($(#1} - 1))
  declare -ir LFBFL_i_max=$(
```

```
159       min "${LFBFL_sort_command}" "${LFBFL_length_minus_1}" "$2"
160     )
161     declare -i LFBFL_i
162     declare -i LFBFL_j
163     local LFBFL_current_char
164
165     # At the beginning, each position has score 0.
166     for ((LFBFL_i=0; LFBFL_i<=LFBFL_i_max; ++LFBFL_i)) do
167       LFBFL_positions["${LFBFL_i}"]="0"
168     done
169     LFBFL_j=$((LFBFL_i+1))
170     LFBFL_positions["${LFBFL_j}"]="0"
171
172     # If we have a forbidden previous character,
173     # we mark the forbidden positions with -1.
174     if [[ -n "$5" ]]; then
175       for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
176         LFBFL_j=$((LFBFL_i+1))
177         LFBFL_current_char="${1:${LFBFL_i}:1}"
178         if [[ "${LFBFL_current_char}" == "$5" ]]; then
179           LFBFL_positions["${LFBFL_j}"]="-1"
180         fi
181       done
182     fi
183
184     if [[ "$4" == "7" ]]; then
185       for ((LFBFL_j=LFBFL_i_max; LFBFL_j>0; --LFBFL_j)) do
186         LFBFL_i=$((LFBFL_j-1))
187         LFBFL_current_char="${1:${LFBFL_i}:1}"
188         eval "$3 '${LFBFL_current_char}' 0"
189         # echo "${split_score_result}|${LFBFL_i}"
190         if [[ split_score_result -ge 1 ]]; then
191           # echo "${split_score_result}|${LFBFL_i}"
192           if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
193             LFBFL_positions["${LFBFL_i}"]=$(
194               max "${LFBFL_sort_command}"\
195                 "${LFBFL_positions["${LFBFL_i}"]}"\
196                 "${split_score_result}"
197             )
198           fi
199         fi
200         eval "$3 '${LFBFL_current_char}' 1"
201         # echo "${split_score_result}|${LFBFL_j}"
202         if [[ split_score_result -ge 1 ]]; then
203           # echo "${split_score_result}|${LFBFL_j}"
204           if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
```

```
205           LFBFL_positions["${LFBFL_j}"]=$(
206             max "${LFBFL_sort_command}"\
207               "${LFBFL_positions["${LFBFL_j}"]}"\
208               "${split_score_result}"
209           )
210           break
211         fi
212       fi
213     done
214   else
215     for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
216       LFBFL_j=$((LFBFL_i+1))
217       LFBFL_current_char="${1:${LFBFL_i}:1}"
218       eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
219       # echo "${split_score_result}|${LFBFL_i}"
220       if [[ split_score_result -ge 1 ]]; then
221         # echo "${split_score_result}|${LFBFL_i}"
222         if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
223           LFBFL_positions["${LFBFL_i}"]=$(
224             max "${LFBFL_sort_command}"\
225               "${LFBFL_positions["${LFBFL_i}"]}"\
226               "${split_score_result}"
227           )
228         fi
229       fi
230       eval "$3 '${LFBFL_current_char}' ${LFBFL_j} 1"
231       # echo "${split_score_result}|${LFBFL_j}"
232       if [[ split_score_result -ge 1 ]]; then
233         # echo "${split_score_result}|${LFBFL_j}"
234         if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
235           LFBFL_positions["${LFBFL_j}"]=$(
236             max "${LFBFL_sort_command}"\
237               "${LFBFL_positions["${LFBFL_j}"]}"\
238               "${split_score_result}"
239           )
240         fi
241       fi
242     done
243   fi
244   declare -ir LFBFL_max_score=$(
245     max 'sort --numeric-sort' "${LFBFL_positions[@]}"
246   )
247   local LFBFL_value
248   declare -i LFBFL_best_position
249   for ((LFBFL_i=0; LFBFL_i<=LFBFL_i_max; ++LFBFL_i)) do
250     local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
```

```
    # echo "${LFBFL_i} ${LFBFL_value}"
    if [[ "${LFBFL_value}" == "${LFBFL_max_score}" ]]; then
      LFBFL_best_position=$((LFBFL_i))
    fi
  done
  # shellcheck disable=SC2248
  split_line_at "$1" ${LFBFL_best_position}
  split_line_at_most_result_start="${split_line_at_result_beginning}"
  split_line_at_most_result_end="${split_line_at_result_end}"
  # echo "$split_line_at_most_result_start"
  # echo "$split_line_at_most_result_end"
}

split_last_line(){
  # $1=$new_lines
  # $2=$prefix
  # $3=$max_length
  # $4=$suffix
  # $5=$split_score_command
  # $6=$split_score_command_properties
  # $7=$forbidden_previous_character ('\' usually)
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
  # This function adapts the effective max position for the split by
  # taking into account the length of the suffix.
  if [[ ! -o pipefail ]]; then
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi
```

```
297
298   declare -g split_last_line_result="$1"
299   declare -ir LFBFL_max_length_plus=$(($3 + 1))
300   declare -ir LFBFL_length2=$(($3 - ${#4}))
301   declare -r LFBFL_regexp='.\{'"${LFBFL_max_length_plus}"'\}$'
302   if
303     echo "$1"\
304       | sed -e 's/\\n/\n/g'\
305       | grep -q "${LFBFL_regexp}";
306   then
307     declare -r LFBFL_start=$(
308       echo "$1"\
309       | sed -e 's/\\n/\n/g'\
310       | head --lines=-1\
311       | sed -z 's/\n/\\n/g'
312     )
313     # echo "start: $LFBFL_start"
314     declare -r LFBFL_last_line=$(
315       echo "$1"\
316       | sed -e 's/\\n/\n/g'\
317       | tail --lines=1
318     )
319     # echo "last_line: ${LFBFL_last_line}"
320     split_last_line_result=""
321     if [[ -n "${LFBFL_start}" ]]; then
322       split_last_line_result="${LFBFL_start}\n"
323     fi
324     if [[ -n "$5" ]]; then
325       split_line_at_most "${LFBFL_last_line}" "${LFBFL_length2}"\
326         "$5" "$6" "$7"
327       split_last_line_result+="${split_line_at_most_result_start}$4"
328       split_last_line_result+="\n"
329       split_last_line_result+="$2${split_line_at_most_result_end}"
330     else
331       split_last_line_result+="${LFBFL_last_line:0:${LFBFL_length2}}"
332       split_last_line_result+="$4\n"
333       split_last_line_result+="$2${LFBFL_last_line:${LFBFL_length2}}"
334     fi
335   fi
336 }
337
338 repeated_split_last_line(){
339   declare -g repeated_split_last_line_result="$1"
340   local LFBFL_result="$1x"
341   until
342     [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
```

```
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # echo "${repeated_split_last_line_result}"
  done
}
```

2.27 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
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#
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

too_long_code_lines(){
```

```
36    # Options:
37    #   --verbose
38    #   --work-directory=""
39    #   --max-line-length
40    declare -i LFBFL_i_verbose=0
41    get_verbose_option "$@"
42    local LFBFL_work_directory=""
43    get_work_directory_option "$@"
44    pushd_to_work_directory\
45      && trap 'popd_from_work_directory' RETURN
46    can_continue_after_enhanced_pushd || return
47
48    local LFBFL_arg
49    declare -i LFBFL_max_line_length=70
50    for LFBFL_arg in "$@"; do
51      if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
52        LFBFL_arg=${LFBFL_arg#--max-line-length=}
53        LFBFL_max_line_length=$((LFBFL_arg))
54        break
55      fi
56    done
57
58    if [[ ! -o pipefail ]]; then
59      [[ LFBFL_i_verbose -eq 1 ]] && echo "pipefail option activated"
60      set -o pipefail
61      trap 'set +o pipefail' RETURN
62    fi
63
64    get_COMMON_TEXT_FILES_GLOB_PATTERNS
65    local LFBFL_pattern
66    local LFBFL_long_line
67    local LFBFL_file_name
68    local LFBFL_line
69    local LFBFL_extension
70    local LFBFL_base_name
71    declare -ir LFBFL_overlength=$((LFBFL_max_line_length+1))
72    for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
73      [[ LFBFL_i_verbose -eq 0 ]]\
74        || echo "Iterating on pattern: ${LFBFL_pattern}"
75      # shellcheck disable=SC2248
76      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
77        | relevant_find\
78        | not_license_find\
79        | xargs grep -H '.\{'${LFBFL_overlength}'\}'\
80        | while read -r LFBFL_long_line;
81      do
```

```
      LFBFL_file_name=${LFBFL_long_line%%:*} # Drop long ':*' suffix
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_name##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_name%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.tpl"
        then
          continue
        fi
        echo "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if ! [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
          echo "${LFBFL_long_line}"
        fi
      else
        echo "${LFBFL_long_line}"
      fi
    done
  done
}
```

2.28 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

source "overwrite_if_not_equal.libr.sh"

update_common_build_and_checks(){
  declare -i LFBFL_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose=1
  fi
  readonly LFBFL_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_verbose -eq 1 ]] && echo "pipefail option activated"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  cp "./${LFBFL_common_file_name}.tpl"\
    "./${LFBFL_common_file_name}.temp"

  declare -ar LFBFL_file_names=(
    "build_md_from_printable_md.exec.sh"
    "check_shell_scripts_beginnings.libr.sh"
    "check_shell_scripts_indentation.libr.sh"
    "check_URLs.libr.sh"
    "common_options.libr.sh"
    "comparisons.libr.sh"
    "create_PDF.exec.sh"
    "generate_from_template.libr.sh"
    "get_common_text_glob_patterns.libr.sh"
    "grammar_and_spell_check.libr.sh"
    "licenses_templates/build_licenses_templates.exec.sh"
    "licenses_templates/license_file_header_AGPLv3+.tpl"
    "licenses_templates/license_file_header_GPLv3+.tpl"
    "licenses_templates/license_file_header_LGPLv3+.tpl"
    "lines_counts.libr.sh"
    "lines_filters.libr.sh"
    "lines_maps.libr.sh"
```

```
    "update_or_check_files_names_listing.exec.sh"
    "overwrite_if_not_equal.libr.sh"
    "python_black_complement.libr.sh"
    "python_isort_complement.libr.sh"
    "split_score.exec.php"
    "split_score.libr.php"
    "strings_functions.libr.sh"
    "too_long_code_lines.libr.sh"
  )

  local LFBFL_file_name
  local LFBFL_file_sha512
  local LFBFL_file_sha512_all
  local LFBFL_base_file_name
  for LFBFL_file_name in "${LFBFL_file_names[@]}"; do
    LFBFL_file_sha512=$(
      sha512sum "./${LFBFL_file_name}"\
      | cut -f1 -d' '
    )
    LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    sed -i\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut -f1 -d' '
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
```

```
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed -i\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

2.29 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
```

```
32 # to "update_or_check_files_names_listing.sh"?
33
34 LFBFL_subdir="build_and_checks_dependencies"
35 # shellcheck source=common_options.libr.sh
36 source "./${LFBFL_subdir}/common_options.libr.sh"
37 # shellcheck source=lines_filters.libr.sh
38 source "./${LFBFL_subdir}/lines_filters.libr.sh"
39 # shellcheck source=strings_functions.libr.sh
40 source "./${LFBFL_subdir}/strings_functions.libr.sh"
41
42 update_or_check_files_names_listing(){
43   # Options:
44   #   --verbose
45   #   --work-directory=""
46   declare -i LFBFL_i_verbose=0
47   get_verbose_option "$@"
48   local LFBFL_work_directory=""
49   get_work_directory_option "$@"
50   pushd_to_work_directory\
51     && trap 'popd_from_work_directory' RETURN
52   can_continue_after_enhanced_pushd || return
53
54   declare -i LFBFL_write=0
55   if [[ "$*" == *--write* ]]; then
56     LFBFL_write=1
57   fi
58   readonly LFBFL_write
59
60   declare -i LFBFL_append=0
61   if [[ "$*" == *--append* ]]; then
62     LFBFL_append=1
63   fi
64   readonly LFBFL_append
65
66   if [[ ! -o pipefail ]]; then
67     [[ LFBFL_i_verbose -eq 1 ]] && echo "pipefail option activated"
68     set -o pipefail
69     trap 'set +o pipefail' RETURN
70   fi
71
72   LFBFL_subdir2="build_and_checks_variables"
73   local LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
74   readonly LFBFL_listing
75   local LFBFL_data_file_name="./${LFBFL_subdir2}/repository_data.txt"
76   readonly LFBFL_data_file_name
77
```

```
  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  if [[ LFBFL_write -eq 1 ]]; then
    : > "${LFBFL_listing}"
  fi
  # Remove line returns here to keep lines short.
  sed -Ez 's/\\\n//Mg' "${LFBFL_listing}"\
    > "${LFBFL_listing}.temp"
  shopt -s dotglob
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_max_line_length} /
  declare -r LFBFL_split_score_command="${get_split_score_result}"
  local LFBFL_split_score_command_properties
  LFBFL_split_score_command_properties="${get_split_score_result2}"
  readonly LFBFL_split_score_command_properties
  local LFBFL_suffix=\\\\ # instead of '\\' to avoid shellcheck SC1003
  readonly LFBFL_suffix
  local LFBFL_file_name
  local LFBFL_base_file_name
  local LFBFL_split_file_name
  find . -type f -printf '%P\n'\
    | relevant_find\
    | sort\
    | while read -r LFBFL_file_name;
  do
    git check-ignore -q "${LFBFL_file_name}" && continue
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
      || continue
    if [[ "${LFBFL_base_file_name}" == *.md ]]; then
      if [[ -f "${LFBFL_file_name}.tpl" ]]; then
        continue
      fi
    fi
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_file_name}"\
      ""\
      ${LFBFL_max_line_length}\
      "${LFBFL_suffix}"\
      "${LFBFL_split_score_command}"\
      "${LFBFL_split_score_command_properties}"
    LFBFL_split_file_name="${repeated_split_last_line_result}"
    if [[ LFBFL_write -eq 1 ]]; then
      # shellcheck disable=SC2001
```

```
      echo "${LFBFL_file_name}"\
        | sed -e "s|${LFBFL_file_name}|${LFBFL_split_file_name}|g"\
        >> "${LFBFL_listing}"
      continue
    fi
    if grep -q "${LFBFL_file_name}\$" "${LFBFL_listing}.temp"; then
      :
    else
      echo\
      "The file ${LFBFL_file_name} is not listed in ${LFBFL_listing}."
      if [[ LFBFL_append -eq 1 ]]; then
        # shellcheck disable=SC2001
        echo "${LFBFL_file_name}"\
          | sed -e "s|${LFBFL_file_name}|${LFBFL_split_file_name}|g"\
          >> "${LFBFL_listing}"
      fi
    fi
  done

  while read -r LFBFL_file_name;
  do
    [[ "${LFBFL_file_name}" != '//'* ]] || continue
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    if ! [[ -f "${LFBFL_file_name}" ]]; then
      echo\
      "The non-file ${LFBFL_file_name} is listed in ${LFBFL_listing}."
    fi
  done < "${LFBFL_listing}.temp"

  rm "${LFBFL_listing}.temp"
  shopt -u dotglob
  shopt -s globstar

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    cat "${LFBFL_listing}"
  fi
}

update_or_check_files_names_listing "$@"
```

2.30 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts

```

2.31 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

2.32 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

2.33 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

2.34 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
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-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
```

```
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
```

```
77  }
78  ol.level2{
79    counter-reset: level2;
80  }
81  ol.level1 li:before{
82    content: counter(level1) ". ";
83    counter-increment: level1;
84  }
85  ol.level2 li:before{
86    content: counter(level1) "." counter(level2) " ";
87    counter-increment: level2;
88  }
89  </style>
90  <title>@repository_name@</title>
91  </head>
92  <body >
93  <table class="title">
94    <tr>
95      <td style="padding:1ex">
96        <h1 class="titlemain">@repository_name@</h1>
97        <h2 class="titlerest">
98          @author_full_name@
99          <br>
100         <a href="@author_website@">
101           <span style="font-family:monospace">
102           @author_website@
103           </span>
104         </a>
105         <br>
106         <a href="mailto:@author_email@">
107           <span style="font-family:monospace">
108           @author_email@
109           </span>
110         </a>
111       </h2>
112     </td>
113   </tr>
114 </table>
115
116 <blockquote class="abstract">
117   <span style="font-weight:bold">Abstract:</span>
118   @abstract@
119 </blockquote>
120 <p>Current version: @current_date@</p>
121 <p>Current number of commits: @number_of_commits@</p>
122 <p>Current git SHA1: @current_git_SHA1@</p>
```

```
<p>
Code lines: @number_of_lines@
</p>


<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files Tree</a></li>
  <li><a href="#section2">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Listing of files</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
The text of the licenses is available at:
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  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
```

```
>@repository_git_url@</span>.
</p>

<div>
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

<p>
Bash can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/bash/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/bash/</span></a>,<br>
<a href="https://tiswww.case.edu/php/chet/bash/bashtop.html"><span
  style="font-family:monospace"
>https://tiswww.case.edu/php/chet/bash/bashtop.html</span></a>,<br>
```

```
<a href="https://savannah.gnu.org/projects/bash/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/bash/</span></a>.
</p>

<p>
GNU coreutils can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/coreutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/coreutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/coreutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/coreutils/</span></a>.
</p>

<p>
uutils coreutils can be found at the following URLs:<br>
<a href="https://uutils.github.io/coreutils/"><span
  style="font-family:monospace"
>https://uutils.github.io/coreutils/</span></a>,<br>
<a href="https://github.com/uutils/coreutils"><span
  style="font-family:monospace"
>https://github.com/uutils/coreutils</span></a>.
</p>

<p>
diff (GNU diffutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/diffutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/diffutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/diffutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/diffutils/</span></a>.
</p>

<p>
find, xargs (GNU findutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/findutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/findutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/findutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/findutils/</span></a>.
</p>

<p>
```

```
grep (GNU grep) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/grep/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/grep/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/grep/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/grep/</span></a>.
</p>

<p>
sed (GNU sed) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/sed/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/sed/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/sed/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/sed/</span></a>.
</p>

<p>
ShellCheck can be found at the following URLs:<br>
<a href="https://www.shellcheck.net/"><span
  style="font-family:monospace"
>https://www.shellcheck.net/</span></a>,<br>
<a href="https://github.com/koalaman/shellcheck"><span
  style="font-family:monospace"
>https://github.com/koalaman/shellcheck</span></a>.
</p>

<p>
GNU Wget can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/wget/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/wget/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/wget/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/wget/</span></a>.
</p>

<p>
more of util-linux can be found at the following URLs:<br>
<a href="https://www.kernel.org/pub/linux/utils/util-linux/"><span
  style="font-family:monospace"
>https://www.kernel.org/pub/linux/utils/util-linux/</span></a>,<br>
<a
href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
```

```
><span
  style="font-family:monospace"
>https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
></a>,<br>
<a href="https://github.com/util-linux/util-linux"><span
  style="font-family:monospace"
>https://github.com/util-linux/util-linux</span></a> official?
</p>

<p>
pandoc can be found at the following URLs:<br>
<a href="https://pandoc.org/"><span
  style="font-family:monospace"
>https://pandoc.org/</span></a>,<br>
<a href="https://github.com/jgm/pandoc"><span
  style="font-family:monospace"
>https://github.com/jgm/pandoc</span></a>.
</p>

<p>
pdftex/pdflatex can be found at the following URLs:<br>
<a href="https://www.tug.org/applications/pdftex/"><span
  style="font-family:monospace"
>https://www.tug.org/applications/pdftex/</span></a>,<br>
<a href="https://tug.org/texlive/svn/"><span
  style="font-family:monospace"
>https://tug.org/texlive/svn/</span></a>.
</p>

<p>
tree can be found at the following URLs:<br>
<a href="https://oldmanprogrammer.net/source.php?dir=projects/tree"
><span style="font-family:monospace"
>https://oldmanprogrammer.net/source.php?dir=projects/tree</span></a>,<br>
<a href="https://gitlab.com/OldManProgrammer/unix-tree"><span
  style="font-family:monospace"
>https://gitlab.com/OldManProgrammer/unix-tree</span></a>,<br>
<a href="https://github.com/Old-Man-Programmer/tree"><span
  style="font-family:monospace"
>https://github.com/Old-Man-Programmer/tree</span></a>.
</p>

<p>
git can be found at the following URLs:<br>
<a href="https://git-scm.com/about"><span
  style="font-family:monospace"
```

```
>https://git-scm.com/about</span></a>,<br>
<a href="https://github.com/git/git"><span
  style="font-family:monospace"
>https://github.com/git/git</span></a>.
</p>

<p>
pylint can be found at the following URLs:<br>
<a href="https://www.pylint.org/"><span
  style="font-family:monospace"
>https://www.pylint.org/</span></a>,<br>
<a href="https://pylint.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://pylint.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/pylint-dev/pylint"><span
  style="font-family:monospace"
>https://github.com/pylint-dev/pylint</span></a>,<br>
<a href="https://pypi.org/project/pylint/"><span
  style="font-family:monospace"
>https://pypi.org/project/pylint/</span></a>.
</p>

<p>
notify-send can be found at the following URLs:<br>
<a href="https://gitlab.gnome.org/GNOME/libnotify/"><span
  style="font-family:monospace"
>https://gitlab.gnome.org/GNOME/libnotify/</span></a>,<br>
<a href="https://gnome.pages.gitlab.gnome.org/libnotify/"><span
  style="font-family:monospace"
>https://gnome.pages.gitlab.gnome.org/libnotify/</span></a>.
</p>

<p>
black can be found at the following URLs:<br>
<a href="https://github.com/psf/black"><span
  style="font-family:monospace"
>https://github.com/psf/black</span></a>,<br>
<a href="https://black.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://black.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://pypi.org/project/black/"><span
  style="font-family:monospace"
>https://pypi.org/project/black/</span></a>.
</p>

<p>
```

```
isort can be found at the following URLs:<br>
<a href="https://pycqa.github.io/isort/"><span
  style="font-family:monospace"
>https://pycqa.github.io/isort/</span></a>,<br>
<a href="https://github.com/pycqa/isort/"><span
  style="font-family:monospace"
>https://github.com/pycqa/isort/</span></a>,<br>
<a href="https://pypi.org/project/isort/"><span
  style="font-family:monospace"
>https://pypi.org/project/isort/</span></a>.
</p>

<p>
mypy can be found at the following URLs:<br>
<a href="https://mypy-lang.org/"><span
  style="font-family:monospace"
>https://mypy-lang.org/</span></a>,<br>
<a href="https://mypy.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://mypy.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/python/mypy"><span
  style="font-family:monospace"
>https://github.com/python/mypy</span></a>,<br>
<a href="https://pypi.org/project/mypy/"><span
  style="font-family:monospace"
>https://pypi.org/project/mypy/</span></a>.
</p>

<p>
ruff can be found at the following URLs:<br>
<a href="https://docs.astral.sh/ruff/"><span
  style="font-family:monospace"
>https://docs.astral.sh/ruff/</span></a>,<br>
<a href="https://github.com/astral-sh/ruff"><span
  style="font-family:monospace"
>https://github.com/astral-sh/ruff</span></a>,<br>
<a href="https://pypi.org/project/ruff/"><span
  style="font-family:monospace"
>https://pypi.org/project/ruff/</span></a>.
</p>

<p>
bandit can be found at the following URLs:<br>
<a href="https://bandit.readthedocs.io/en/latest/"><span
  style="font-family:monospace"
>https://bandit.readthedocs.io/en/latest/</span></a>,<br>
```

```
<a href="https://github.com/PyCQA/bandit"><span
  style="font-family:monospace"
>https://github.com/PyCQA/bandit</span></a>,<br>
<a href="https://pypi.org/project/bandit/"><span
  style="font-family:monospace"
>https://pypi.org/project/bandit/</span></a>.
</p>

<p>
phpmd can be found at the following URLs:<br>
<a href="https://phpmd.org/"><span
  style="font-family:monospace"
>https://phpmd.org/</span></a>,<br>
<a href="https://github.com/phpmd/phpmd"><span
  style="font-family:monospace"
>https://github.com/phpmd/phpmd</span></a>,<br>
<a href="https://packagist.org/packages/phpmd/phpmd"><span
  style="font-family:monospace"
>https://packagist.org/packages/phpmd/phpmd</span></a>.
</p>

<p>
php can be found at the following URLs:<br>
<a href="https://www.php.net/"><span
  style="font-family:monospace"
>https://www.php.net/</span></a>,<br>
<a href="https://github.com/php/php-src"><span
  style="font-family:monospace"
>https://github.com/php/php-src</span></a>.
</p>

<p>
tex can be found at the following URL:<br>
<a href="https://ctan.org/tex-archive/systems/knuth/dist/tex"><span
  style="font-family:monospace"
>https://ctan.org/tex-archive/systems/knuth/dist/tex</span></a>.
</p>

<p>
pcregrep, pcre2grep can be found at the following URLs:<br>
<a href="https://www.pcre.org/"><span
  style="font-family:monospace"
>https://www.pcre.org/</span></a>,<br>
<a href="https://pcre2project.github.io/pcre2/"><span
  style="font-family:monospace"
>https://pcre2project.github.io/pcre2/</span></a>,<br>
```

```
491 <a href="https://github.com/PCRE2Project/pcre1"><span
492   style="font-family:monospace"
493 >https://github.com/PCRE2Project/pcre1</span></a>,<br>
494 <a href="https://github.com/PCRE2Project/pcre2"><span
495   style="font-family:monospace"
496 >https://github.com/PCRE2Project/pcre2</span></a>,<br>
497 <a
498 href="https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
499 ><span
500   style="font-family:monospace"
501 >https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c</span
502 ></a>.
503 </p>
504
505 <p>
506 If you see something that I missed regarding my dependencies,
507 please tell me/email me :).
508 </p>
509
510 @files_listing_HTMLPreInput@
511
512 <p>
513 @acknowledgments@
514 </p>
515
516 <script>
517 "use strict";
518 const pres = Array.from(
519   document.getElementsByClassName('numbered_lines')
520 );
521 pres.forEach(
522   (pre) => {
523     const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
524     let numbers = "";
525     for(let i = 1; i <= nb; ++i){
526       numbers += i.toString();
527       if(i < nb){ numbers += "\n "; }
528     }
529     pre.setAttribute('line-numbers', numbers);
530   }
531 );
532 </script>
533 </html>
```

2.35 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
```

```
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files tree}
\label{section:tree}

\begin{verbatim}
```

```
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Listing of files}
\label{section:listing}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
\end{itemize}

```

```
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:\newline
\url{https://www.gnu.org/software/bash/},\newline
\url{https://tiswww.case.edu/php/chet/bash/bashtop.html},\newline
\url{https://savannah.gnu.org/projects/bash/}.

GNU coreutils can be found at the following URLs:\newline
\url{https://www.gnu.org/software/coreutils/},\newline
\url{https://savannah.gnu.org/projects/coreutils/}.

uutils coreutils can be found at the following URLs:\newline
\url{https://uutils.github.io/coreutils/},\newline
\url{https://github.com/uutils/coreutils}.

diff (GNU diffutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/diffutils/},\newline
\url{https://savannah.gnu.org/projects/diffutils/}.

find, xargs (GNU findutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/findutils/},\newline
\url{https://savannah.gnu.org/projects/findutils/}.

grep (GNU grep) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/grep/},\newline
\url{https://savannah.gnu.org/projects/grep/}.

sed (GNU sed) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/sed/},\newline
\url{https://savannah.gnu.org/projects/sed/}.

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\url{https://www.shellcheck.net/},\newline
\url{https://github.com/koalaman/shellcheck}.

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\url{https://savannah.gnu.org/projects/wget/}.

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{\small
  \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
```

```
},
\newline
\url{https://github.com/util-linux/util-linux} official?

pandoc can be found at the following URLs:\newline
\url{https://pandoc.org/},\newline
\url{https://github.com/jgm/pandoc}.

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\url{https://tug.org/texlive/svn/}.

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\url{https://oldmanprogrammer.net/source.php?dir=projects/tree},\newline
\url{https://gitlab.com/OldManProgrammer/unix-tree},\newline
\url{https://github.com/Old-Man-Programmer/tree}.

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\url{https://github.com/git/git}.

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\url{https://www.pylint.org/},\newline
\url{https://pylint.readthedocs.io/en/stable/},\newline
\url{https://github.com/pylint-dev/pylint},\newline
\url{https://pypi.org/project/pylint/}.

notify-send can be found at the following URLs:\newline
\url{https://gitlab.gnome.org/GNOME/libnotify/},\newline
\url{https://gnome.pages.gitlab.gnome.org/libnotify/}.

black can be found at the following URLs:\newline
\url{https://github.com/psf/black},\newline
\url{https://black.readthedocs.io/en/stable/},\newline
\url{https://pypi.org/project/black/}.

isort can be found at the following URLs:\newline
\url{https://pycqa.github.io/isort/},\newline
\url{https://github.com/pycqa/isort/},\newline
\url{https://pypi.org/project/isort/}.

mypy can be found at the following URLs:\newline
\url{https://mypy-lang.org/},\newline
\url{https://mypy.readthedocs.io/en/stable/},\newline
\url{https://github.com/python/mypy},\newline
\url{https://pypi.org/project/mypy/}.
```

```

ruff can be found at the following URLs:\newline
\url{https://docs.astral.sh/ruff/},\newline
\url{https://github.com/astral-sh/ruff},\newline
\url{https://pypi.org/project/ruff/}.

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\url{https://bandit.readthedocs.io/en/latest/},\newline
\url{https://github.com/PyCQA/bandit},\newline
\url{https://pypi.org/project/bandit/}.

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\url{https://phpmd.org/},\newline
\url{https://github.com/phpmd/phpmd},\newline
\url{https://packagist.org/packages/phpmd/phpmd}.

php can be found at the following URLs:\newline
\url{https://www.php.net/},\newline
\url{https://github.com/php/php-src}.

tex can be found at the following URL:\newline
\url{https://ctan.org/tex-archive/systems/knuth/dist/tex}.

pcregrep, pcre2grep can be found at the following URLs:\newline
\url{https://www.pcre.org/},\newline
\url{https://pcre2project.github.io/pcre2/},\newline
\url{https://github.com/PCRE2Project/pcre1},\newline
\url{https://github.com/PCRE2Project/pcre2},\newline
{\small
  \url{https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c}
}.

If you see something that I missed regarding my dependencies,
please tell me/email me :).

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

2.36 build_and_checks_variables/files_listing.tex.tpl

```
\subsection{
  build\_and\_checks\_dependencies/build\_and\_checks.exec.sh.tpl
}
```

```
\label{
  build_and_checks_dependencies:build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/build\_md\_from\_printable\_md.exec.sh
}
\label{
  build_and_checks_dependencies:build_md_from_printable_mdexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_beginnings.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_beginningslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_indentation.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_indentationlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
}

```

```
\subsection{
  build\_and\_checks\_dependencies/check\_URLs.libr.sh
}
\label{
  build_and_checks_dependencies:check_URLslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_URLs.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/common\_options.libr.sh
}
\label{
  build_and_checks_dependencies:common_optionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_options.libr.sh
```

```
}


\subsection{
  build\_and\_checks\_dependencies/comparisons.libr.sh
}
\label{
  build_and_checks_dependencies:comparisonslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/comparisons.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/create\_PDF.exec.sh
}
\label{
  build_and_checks_dependencies:create_PDFexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/create_PDF.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/generate\_from\_template.libr.sh
}
\label{
  build_and_checks_dependencies:generate_from_templatelibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/generate_from_template.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
get\_common\_text\_glob\_patterns.libr.sh
}
\label{
  build_and_checks_dependencies:get_common_text_glob_patternslibrsh
}
```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/grammar\_and\_spell\_check.libr.sh
}
\label{
  build_and_checks_dependencies:grammar_and_spell_checklibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
build\_licenses\_templates.exec.sh
}
\label{
  build_and_checks_dependencies:licenses_templates:%
build_licenses_templatesexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
build_licenses_templates.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_AGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_AGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_AGPLv3+.tpl
}
```

```
\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_GPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_GPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_GPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_LGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_LGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_LGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/lines\_counts.libr.sh
}
\label{
  build_and_checks_dependencies:lines_countslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_counts.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_filters.libr.sh
```

```
}
\label{
  build_and_checks_dependencies:lines_filterslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_filters.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_maps.libr.sh
}
\label{
  build_and_checks_dependencies:lines_mapslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_maps.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/overwrite\_if\_not\_equal.libr.sh
}
\label{
  build_and_checks_dependencies:overwrite_if_not_equallibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_black\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_black_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_black_complement.libr.sh
}
```

```
\subsection{
  build\_and\_checks\_dependencies/python\_isort\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_isort_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_isort_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.exec.php
}
\label{
  build_and_checks_dependencies:split_scoreexecphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.exec.php
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.libr.php
}
\label{
  build_and_checks_dependencies:split_scorelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.libr.php
}


\subsection{
  build\_and\_checks\_dependencies/strings\_functions.libr.sh
}
\label{
  build_and_checks_dependencies:strings_functionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/strings_functions.libr.sh
}
```

```
\subsection{
  build\_and\_checks\_dependencies/too\_long\_code\_lines.libr.sh
}
\label{
  build_and_checks_dependencies:too_long_code_lineslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/too_long_code_lines.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:update_common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_or\_check\_files\_names\_listing.exec.sh
}
\label{
  build_and_checks_dependencies:update_or_check_files_names_listingexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh
}


\subsection{
  build\_and\_checks\_variables/.gitignore
}
\label{
  build_and_checks_variables:gitignore
}
```

```latex

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/.gitignore
}


\subsection{
  build\_and\_checks\_variables/bandit.ini
}
\label{
  build_and_checks_variables:banditini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit.ini
}


\subsection{
  build\_and\_checks\_variables/bandit\_baseline.json
}
\label{
  build_and_checks_variables:bandit_baselinejson
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit_baseline.json
}


\subsection{
  build\_and\_checks\_variables/black.toml
}
\label{
  build_and_checks_variables:blacktoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/black.toml
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.html.tpl
}
\label{
```

```
  build_and_checks_variables:DevOrSysAdminScriptshtmltpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.tex.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptstextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_listing.tex.tpl
}
\label{
  build_and_checks_variables:files_listingtextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_listing.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_names\_listing.txt
}
\label{
  build_and_checks_variables:files_names_listingtxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_names_listing.txt
}


\subsection{
  build\_and\_checks\_variables/licenses\_notes.txt
```

```
}
\label{
  build_and_checks_variables:licenses_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/licenses_notes.txt
}


\subsection{
  build\_and\_checks\_variables/phpmd\_baseline.xml
}
\label{
  build_and_checks_variables:phpmd_baselinexml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/phpmd_baseline.xml
}


\subsection{
  build\_and\_checks\_variables/pylintrc.toml
}
\label{
  build_and_checks_variables:pylintrctoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/pylintrc.toml
}


\subsection{
  build\_and\_checks\_variables/repository\_data.txt
}
\label{
  build_and_checks_variables:repository_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/repository_data.txt
}
```

```
\subsection{
  build\_and\_checks\_variables/ruff.toml
}
\label{
  build_and_checks_variables:rufftoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/ruff.toml
}


\subsection{
  build\_and\_checks\_variables/shellcheck.ini
}
\label{
  build_and_checks_variables:shellcheckini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck.ini
}


\subsection{
  build\_and\_checks\_variables/shellcheck\_notes.txt
}
\label{
  build_and_checks_variables:shellcheck_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck_notes.txt
}


\subsection{
  build\_and\_checks.exec.sh
}
\label{
  build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks.exec.sh
}
```

```
\subsection{
  COPYING
}
\label{
  COPYING
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING
}


\subsection{
  COPYING.LESSER
}
\label{
  COPYINGLESSER
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING.LESSER
}


\subsection{
  git\_helpers/git\_checks.libr.sh
}
\label{
  git_helpers:git_checkslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/git_checks.libr.sh
}


\subsection{
  git\_helpers/post-commit
}
\label{
  git_helpers:post-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
```

```
git_helpers/post-commit
}


\subsection{
  git\_helpers/post\_commit\_renaming\_comment.libr.sh
}
\label{
  git_helpers:post_commit_renaming_commentlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post_commit_renaming_comment.libr.sh
}


\subsection{
  git\_helpers/pre-commit
}
\label{
  git_helpers:pre-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/pre-commit
}


\subsection{
  IMAP\_transfert.py
}
\label{
  IMAP_transfertpy
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
IMAP_transfert.py
}


\subsection{
  LiensUtiles.txt
}
\label{
  LiensUtilestxt
}
```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
LiensUtiles.txt
}


\subsection{
  README.md.tpl
}
\label{
  READMEmdtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
README.md.tpl
}


\subsection{
  unicode.libr.php
}
\label{
  unicodelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
unicode.libr.php
}


\subsection{
  upgrade\_build\_and\_checks.exec.sh
}
\label{
  upgrade_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
upgrade_build_and_checks.exec.sh
}


\subsection{
  wget\_sha512.libr.sh
}
\label{
```

```
  wget_sha512librsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
wget_sha512.libr.sh
}


```

2.37 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
```

```
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_listing.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
```

```
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
// README.html
// README.md
README.md.tpl
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

2.38 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
```

```
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception.

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
```

```
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/
```

```

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

2.39 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName" file="build_and
  <violation rule="PHPMD\Rule\Naming\LongVariable" file="build_and_checks_dependenc
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName" file="build_and_
  <violation rule="PHPMD\Rule\Design\LongMethod" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName" file="unicode.l
  <violation rule="PHPMD\Rule\Naming\LongVariable" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName" file="unicode.li
  <violation rule="PHPMD\Rule\CleanCode\MissingImport" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\CyclomaticComplexity" file="unicode.libr.php"/>
```

```
  <violation rule="PHPMD\Rule\Design\NpathComplexity" file="unicode.libr.php"/>
</phpmd-baseline>
```

2.40 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

2.41 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

```

```
author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

grammar_or_spell_checker_command=more


Why one venv for each Python dependency ?
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
```

```
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
 "A":{
  "1.1.1": {
    "B": {
      "2.2.2": {
        "C": {
           // Dumb, you did not found it by yourself? Failed exam.
          "3.3.3" : {}
        }
      }
    },
    "D": {
      "0.1.0": {
        "C": {
           // Dumb, you did not found it by yourself? Failed exam.
          "4.4.4" : {}
        }
      }
    }
  },
  "B": {
    "2.2.2": {
      "C": {
        // Dumb, you did not found it by yourself? Failed exam.
        "4.4.4" : {}
      }
    }
  }
}
Wonderful someone rediscovered nested structures XD
Seriously so many engineers or computer scientists are dumb to cry.

bandit_venv=../venv_bandit
black_venv=../venv_black
isort_venv=../venv_isort
mypy_venv=../venv_mypy
pylint_venv=../venv_pylint
ruff_venv=../venv_ruff


upgrade venvs can be auto or ask, any other value is interpreted
as ask.
```

```

upgrade_venvs=ask


time interval can be 0 to always upgrade or ask to upgrade,
or a number at most bash max int
https://superuser.com/questions/1030122/
what-is-the-maximum-value-of-a-numeric-bash-shell-variable
it should be enough to simulate never "upgrade or ask for upgrade"
unless you claim to live many billions years ;)
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800
```

2.42 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
```

```
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

2.43 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"

```

2.44 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck:
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
```

```
            source venv/bin/active
- SC1091 : Not following (file not found most of the time).
            Only remaining cases are in common_build_and_checks
            when sourcing some Python venv
            source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
            False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
            False positive when c contains \n.
- SC2034 : Unused variable
            False positive for globals that are declared in functions
            that are not currently called.
            Somehow it can be used to detect dead code.
- SC2125 : Brace expansions and globs are literal.
            False positive when assigning a variable declared with -i.
            https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
            False positive when using "$@" line return and pipes.
            https://github.com/koalaman/shellcheck/issues/3192
            Example:
              # shellcheck disable=SC2145
              all_self_code_lines "$@"\
                | empty_lines_after_file_name
- SC2162 : read without -r will mangle backslashes.
            False positive when read is used without variable:
            https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
            False positive on returning integer values.
            https://github.com/koalaman/shellcheck/issues/3405
- SC2312 : Some command is not invoked separately and thus its return
            value is masked. Frequent with pipes.

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2034$" -e "=SC2125$" -e "=SC2145$" -e "=SC2162$"\
  -e "=SC2248$" -e "=SC2312$" -e "=SC2162,SC2312$"
```

2.45 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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#
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#
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
```

```
53      {
54        echo "temp/"
55        echo ""
56      } > "${LFBFL_variables_directory}/.gitignore"
57    fi
58
59    # LFBFL_dependencies_raw_content_download_URL
60    local LFBFL_dependencies_URL
61    LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
62    LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
63    readonly LFBFL_dependencies_URL
64
65    declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
66    # LFBFL_script_download_URL
67    declare -r\
68      LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
69    declare -r\
70      LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
71    local LFBFL_correct_sha512
72    LFBFL_correct_sha512='35da10baa8c7f7010d5dd80f986c4dce1ff8cd519fe7f'
73    LFBFL_correct_sha512+='dd73c08ee0c3c77ced02c1d04792f99de055732ae4d5'
74    LFBFL_correct_sha512+='6de75c9e78bf817ab530f79ec33e5bf73310067'
75    wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
76      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
77    chmod +x "./${LFBFL_file_path}"
78
79    "${LFBFL_file_path}" "${LFBFL_work_directory}"\
80      "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
81  }
82
83  if [[ "$*" == *--fixed_point_build* ]]; then
84    echo "--fixed_point_build"
85    source "./build_and_checks_dependencies/lines_filters.libr.sh"
86    LFBFL_work_directory="."
87    if [[ -n "$1" ]]; then
88      LFBFL_work_directory="$1"
89    fi
90    LFBFL_variables_directory="${LFBFL_work_directory}/"
91    LFBFL_variables_directory+="build_and_checks_variables"
92    LFBFL_data_file_name="${LFBFL_variables_directory}/"
93    LFBFL_data_file_name+="repository_data.txt"
94    LFBFL_repository_name=""
95    grep_variable "${LFBFL_data_file_name}" repository_name\
96      --result-variable-prefix="LFBFL_"\
97      --replace-line-returns-by=""
98    # Touching the 3 following files first let us iterate
```

```
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed_point_build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.46 COPYING

```
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                       Version 3, 29 June 2007

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```

```

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```

```
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        1) Use a suitable shared library mechanism for linking with the
        Library.  A suitable mechanism is one that (a) uses at run time
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        system, and (b) will operate properly with a modified version
        of the Library that is interface-compatible with the Linked
        Version.

    e) Provide Installation Information, but only if you would otherwise
    be required to provide such information under section 6 of the
    GNU GPL, and only to the extent that such information is
    necessary to install and execute a modified version of the
    Combined Work produced by recombining or relinking the
    Application with a modified version of the Linked Version. (If
    you use option 4d0, the Installation Information must accompany
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```

```

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```

2.48 git_helpers/git_checks.libr.sh

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# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
  declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
  local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
  LFBFL_send_body_2+=" dès la création du modèle"
  readonly LFBFL_send_body_2
  declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
  declare -r LFBFL_send_body_3=\
"Les nouveaux preloadings sont interdits -> #Prefetch()"
  git diff --cached --name-only\
    > /tmp/DOSAS_django_git_check1.temp
  if grep models /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
  fi
```

```
59    if grep serializer /tmp/DOSAS_django_git_check1.temp;
60    then
61      notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
62    fi
63    rm /tmp/DOSAS_django_git_check1.temp
64    return 0
65  }
66
67
68  check_no_abusive_trailing_comma(){
69    # Useful for Python code, although it regularly yields
70    # false positives in Python code
71    # (but it is easy to distinguish by hand).
72    # Use `git commit --no-verify` after check.
73    declare -r LFBFL_send_summary_1="ATTENTION"
74    local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
75    LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
76    readonly LFBFL_send_body_1
77    git diff --cached -r\
78      > /tmp/DOSAS_django_git_check2.temp
79    if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
80    then
81      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
82      return 1
83    fi
84    rm /tmp/DOSAS_django_git_check2.temp
85    return 0
86  }
87
88
89  check_black_code_formatting(){
90    # attention ça ne marche que sur les fichiers "stagés" avec git add
91    declare -r LFBFL_files_string=$(
92      git diff --cached --name-only\
93      | grep '\.py'
94    )
95    echo "${LFBFL_files_string}"
96    declare -a LFBFL_some_files
97    mapfile -t LFBFL_some_files <<< "${LFBFL_files_string}"
98    readonly LFBFL_some_files
99    declare -i LFBFL_error=0
100   # Not very useful since LFBFL_file should be local to the loop by
101   # default.
102   local LFBFL_file
103   for LFBFL_file in "${LFBFL_some_files[@]}"; do
104     echo "Black will check formatting on file ${LFBFL_file}"
```

```
    if ! black --check --diff "${LFBFL_file}";
    then
      LFBFL_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}
```

2.49 git_helpers/post-commit

```
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# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_library="../DevOrSysAdminScripts/git_helpers/"
LFBFL_library+="post_commit_renaming_comment.libr.sh"
# shellcheck source=post_commit_renaming_comment.libr.sh
```

```
source "${LFBFL_library}"
LFBFL_library="../DevOrSysAdminScripts/build_and_checks_dependencies/"
LFBFL_library+="lines_filters.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "${LFBFL_library}"

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

2.50 git_helpers/post_commit_renaming_comment.libr.sh

```
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#
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# 2024-05-31T01:49+02:00: This file was renamed from
```

```
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_cd_result2
pushd .
cd "${LFBFL_some_directory}" || {
  LFBFL_cd_result2=$?
  echo "post_commit_renaming_comment.libr.sh no such directory"
  # shellcheck disable=SC2248
  exit ${LFBFL_cd_result2}
}
# shellcheck source=strings_functions.libr.sh
source "build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_popd_result2
popd || {
  LFBFL_popd_result2=$?
  echo "post_commit_renaming_comment.libr.sh no popd"
  # shellcheck disable=SC2248
  exit ${LFBFL_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# For this reason, this function will only handle when all files
# impacted in the preceding commit are only renamed.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
commit_a_file_renamed_comment(){
  local LFBFL_arg
  declare -i LFBFL_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 commit_a_file_renamed_comment $*"
    LFBFL_verbose=1
```

```
fi
readonly LFBFL_verbose

if [[ ! -o pipefail ]]; then
  [[ LFBFL_verbose -eq 1 ]] && echo "pipefail option activated"
  set -o pipefail
  trap 'set +o pipefail' RETURN
fi

declare -i LFBFL_log_directory_change=0
if [[ "$*" == *--log-directory-change* ]]; then
  LFBFL_log_directory_change=1
fi
readonly LFBFL_log_directory_change

declare -i LFBFL_max_comment_line_length=70
for LFBFL_arg in "$@"; do
  if [[ "${LFBFL_arg}" == --max-comment-line-length=* ]]; then
    LFBFL_arg=${LFBFL_arg#--max-comment-line-length=}
    LFBFL_max_comment_line_length=$((LFBFL_arg))
    break
  fi
done
readonly LFBFL_max_comment_line_length

if [[ LFBFL_verbose -eq 1 ]]; then
  echo "LFBFL_log_directory_change: ${LFBFL_log_directory_change}"
  echo "LFBFL_max_comment_line_length:"\
    "${LFBFL_max_comment_line_length}"
fi
#-----------------------------------------------------------------------

declare -r LFBFL_renaming_lines=$(
  git log -p -1\
  | grep '^diff --git' -A 1
)
if [[ LFBFL_verbose -eq 1 ]]; then
  echo "LFBFL_renaming_lines: ${LFBFL_renaming_lines}"
fi

declare -a LFBFL_renaming_lines_array
mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
readonly LFBFL_renaming_lines_array

declare -r LFBFL_timestamp=$(date --iso-8601="minute")
if [[ LFBFL_verbose -eq 1 ]]; then
```

```
117       echo "LFBFL_timestamp: ${LFBFL_timestamp}"
118     fi
119     get_split_score_simple 1 "${LFBFL_max_comment_line_length}" _
120     LFBFL_split_score_command1="${get_split_score_result}"
121     LFBFL_split_score_command_properties1="${get_split_score_result2}"
122     get_split_score_simple 1 "${LFBFL_max_comment_line_length}" /
123     LFBFL_split_score_command2="${get_split_score_result}"
124     LFBFL_split_score_command_properties2="${get_split_score_result2}"
125     declare -i LFBFL_renaming_happened=0
126     declare -i LFBFL_i
127     declare -i LFBFL_j
128     # declare -r LFBFL_special_file_name="files_names_listing.txt"
129     declare -r LFBFL_sfn="files_names_listing.txt"
130     local LFBFL_diff_line
131     local LFBFL_similarity_line
132     local LFBFL_old_file_path
133     local LFBFL_old_file_name
134     local LFBFL_old_file_directory
135     local LFBFL_new_file_path
136     local LFBFL_new_file_name
137     local LFBFL_new_file_directory
138     local LFBFL_useful_file_name
139     local LFBFL_extension
140     local LFBFL_comment_prefix
141     local LFBFL_comment_prefix2
142     local LFBFL_old_file_name2
143     local LFBFL_old_file_directory2
144     local LFBFL_new_file_name2
145     local LFBFL_new_file_directory2
146     local LFBFL_new_comment
147     declare -i LFBFL_copyright_line_number
148     declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
149     declare -i LFBFL_line_count
150     declare -i LFBFL_lines_after
151     local LFBFL_temp_file_path
152     for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
153       LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
154       LFBFL_i=$((LFBFL_i + 1))
155       LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
156       LFBFL_i=$((LFBFL_i + 1))
157       # We need to increment 3 times since grep adds "--" lines between.
158       LFBFL_i=$((LFBFL_i + 1))
159       if [[ LFBFL_verbose -eq 1 ]]; then
160         echo "LFBFL_diff_line: ${LFBFL_diff_line}"
161         echo "LFBFL_similarity_line: ${LFBFL_similarity_line}"
162       fi
```

```
    if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
      continue
    fi

    # Extract old file name from diff line. --------------------------
    LFBFL_old_file_path=$(
      echo "${LFBFL_diff_line}"\
      | cut -d ' ' -f 3
    )
    # It starts by "a/".
    LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
    LFBFL_old_file_name=$(basename "${LFBFL_old_file_path}")
    LFBFL_old_file_directory=$(dirname "${LFBFL_old_file_path}")
    # Extract new file name from diff line. --------------------------
    LFBFL_new_file_path=$(
      echo "${LFBFL_diff_line}"\
      | cut -d ' ' -f 4
    )
    # It starts by "b/".
    LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
    LFBFL_new_file_name=$(basename "${LFBFL_new_file_path}")
    LFBFL_new_file_directory=$(dirname "${LFBFL_new_file_path}")
    if [[ LFBFL_verbose -eq 1 ]]; then
      echo "LFBFL_old_file_path: ${LFBFL_old_file_path}"
      echo "LFBFL_old_file_name: ${LFBFL_old_file_name}"
      echo "LFBFL_old_file_directory: ${LFBFL_old_file_directory}"
      echo "LFBFL_new_file_path: ${LFBFL_new_file_path}"
      echo "LFBFL_new_file_name: ${LFBFL_new_file_name}"
      echo "LFBFL_new_file_directory: ${LFBFL_new_file_directory}"
    fi
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. ------------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_verbose -eq 1 ]]; then
      echo "LFBFL_useful_file_name: ${LFBFL_useful_file_name}"
      echo "LFBFL_extension: ${LFBFL_extension}"
    fi
    for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_verbose -eq 1 ]]; then
        echo "LFBFL_useful_file_name: ${LFBFL_useful_file_name}"
```

```
        echo "LFBFL_extension: ${LFBFL_extension}"
      fi
    done
    LFBFL_comment_prefix=""
    if [[ "${LFBFL_extension}" == "sh" ]]; then
      LFBFL_comment_prefix="# "
    fi
    if [[ "${LFBFL_extension}" == "sql" ]]; then
      LFBFL_comment_prefix="-- "
    fi
    if [[ "${LFBFL_extension}" == "tex" ]]; then
      LFBFL_comment_prefix="% "
    fi
    if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
      LFBFL_comment_prefix="// "
    fi
    if [[ LFBFL_verbose -eq 1 ]]; then
      echo "LFBFL_comment_prefix: ${LFBFL_comment_prefix}"
    fi
    #-----------------------------------------------------------------
    # Split if necessary old_file_name, new_file_name, etc. to have
    # comment lines below max_comment_line_length.
    LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
    # -----
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command1}"\
      "${LFBFL_split_score_command_properties1}"
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
      "${LFBFL_split_score_command_properties2}"
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command1}"\
      "${LFBFL_split_score_command_properties1}"
```

```
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
      "${LFBFL_split_score_command_properties2}"
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. -----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
    then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from\n"
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to\n"
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_log_directory_change -eq 1 ]] && [[\
      "${LFBFL_new_file_directory}"\
      != "${LFBFL_old_file_directory}"\
    ]]; then
      if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
      then
        LFBFL_new_comment+="\n"
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from\n"
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to\n"
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
    if [[ LFBFL_verbose -eq 1 ]]; then
      echo "LFBFL_new_comment: ${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. -------------------------------------
    LFBFL_copyright_line_number=$(
      grep -n '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut -d ':' -f 1
    )
    if [[ LFBFL_verbose -eq 1 ]]; then
      echo "LFBFL_copyright_line_number:"\
```

```
        "${LFBFL_copyright_line_number}"
    fi
    if [[ LFBFL_copyright_line_number -eq 0 ]]; then
      notify-send "${LFBFL_send_summary_2}"\
        "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
      if [[ LFBFL_verbose -eq 1 ]]; then
        echo "No copyright line. Skipping file."
      fi
      continue
    fi
    # Get total number of lines. -------------------------------------
    LFBFL_line_count=$(
      wc -l\
      < "${LFBFL_new_file_path}"
    )
    if [[ LFBFL_verbose -eq 1 ]]; then
      echo "LFBFL_line_count: ${LFBFL_line_count}"
    fi
    # Compute number of lines after. ---------------------------------
    LFBFL_lines_after=$((
      LFBFL_line_count - LFBFL_copyright_line_number
    ))
    if [[ LFBFL_verbose -eq 1 ]]; then
      echo "LFBFL_lines_after: ${LFBFL_lines_after}"
    fi
    # Update file. ---------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_copyright_line_number}"\
        "${LFBFL_new_file_path}"
      echo -e "${LFBFL_new_comment}"
      tail --lines="${LFBFL_lines_after}" "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. ---------------------------------------------
    git add "${LFBFL_new_file_path}"
    LFBFL_renaming_happened=1
  done
  readonly LFBFL_renaming_happened
  if [[ LFBFL_renaming_happened -eq 1 ]]; then
    # Commit ---------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    git commit --message="${LFBFL_message}"
  fi
```

```
}
```

2.51 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_library="../DevOrSysAdminScripts/git_helpers/"
LFBFL_library+="git_checks.libr.sh"
# shellcheck source=git_checks.libr.sh
source "${LFBFL_library}"

declare -ig LFBFL_git_hook_error=0
if ! check_files; then
  LFBFL_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
```

```
  LFBFL_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_git_hook_error=1
fi
readonly LFBFL_git_hook_error
if [[ LFBFL_git_hook_error -eq 1 ]]; then
  exit 1
fi
exit 0
```

2.52 IMAP_transfert.py

```
"""
This file is part of DevOrSysAdminScripts library.

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Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


```

```
class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don’t see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don’t see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don’t need "pass" XD.
    """


def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
```

```
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
```

```
        if verbose:
            print("Creating new folder in to account.")
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
```

```
172             "Transfert all emails from some IMAP account"
173             " to another IMAP account"
174         ),
175     )
176     parser.add_argument(
177         "from_login",
178         type=str,
179         help="Email account login of user one.",
180     )
181     parser.add_argument(
182         "from_password",
183         type=str,
184         help="Email account password of user one.",
185     )
186     parser.add_argument(
187         "to_login",
188         type=str,
189         help="Email account login of user two.",
190     )
191     parser.add_argument(
192         "to_password",
193         type=str,
194         help="Email account password of user two.",
195     )
196     parser.add_argument(
197         "--to_new_folder",
198         type=str,
199         default="",
200         help="Transfert emails into a new folder of this name.",
201     )
202     parser.add_argument(
203         "--from_host",
204         type=str,
205         default="imap.gmail.com",
206         help="Host of first email account",
207     )
208     parser.add_argument(
209         "--from_port",
210         type=int,
211         default=993,
212         help="Port on host of first email account",
213     )
214     parser.add_argument(
215         "--to_host",
216         type=str,
217         default="imap.gmail.com",
```

```
        help="Host of second email account",
    )
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
            "-------------------------------------------------------\n"
            "IMAP transfert:\n"
            "-------------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-------------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
```

```
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

2.53 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
```

2.54 README.md.tpl

```
# DevOrSysAdminScripts

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[![Codacy-badge]][Codacy-package-page]
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[![Imports:isort:badge]][Isort-GitHub.io]
[![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
[![Linting:pylint:badge]][Pylint-GitHub.com]
[![Linting:ruff:badge]][Ruff-GitHub.com]

```

```
|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.

But the files "pre-commit" and "post-commit" were not renamed,
since it is not possible to give them other names to use them as
pre-commit/post-commit hooks in `git`.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts

[Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
```

```
3dd6ef85e8db4460991d1ebc404b68dd

[Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

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languages/top/llyaudet/DevOrSysAdminScripts

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languages/code-size/llyaudet/DevOrSysAdminScripts

[Security:bandit:badge]: https://img.shields.io/badge/\
security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

2.55 unicode.libr.php

```
<?php
/*
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This file was renamed from "unicode.php" to "unicode.libr.php".
*/

function decimal_code_point_to_UTF8(
  $i_code_point_in_decimal_notation
){
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
  if($i_code_point_in_decimal_notation < 128){
    return chr($i_code_point_in_decimal_notation);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 256 * (2**3)){
    $i_first_byte_base_value = 192;
    $i_last_byte_significant_bits = (
      $i_code_point_in_decimal_notation % (2**6)
    );
    $i_first_byte_significant_bits = intdiv(
      (
```

```
45          $i_code_point_in_decimal_notation
46          - $i_last_byte_significant_bits
47        ),
48        2**6,
49      );
50      $s_result = chr(
51        $i_first_byte_base_value
52        + $i_first_byte_significant_bits
53      );
54      $s_result .= chr(
55        $i_continuation_base_value
56        + $i_last_byte_significant_bits
57      );
58      return $s_result;
59    }
60    // 1110xxxx 10xxxxxx 10xxxxxx
61    if($i_code_point_in_decimal_notation < 256 * 256){
62      $i_first_byte_base_value = 224;
63      $arr_arr_data_per_byte_reverse = [
64        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
65        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
66        ["i_bits" => 4, "i_base_value" => $i_first_byte_base_value],
67      ];
68      $i_rest = $i_code_point_in_decimal_notation;
69      $s_result = "";
70      foreach($arr_arr_data_per_byte_reverse as $arr_data){
71        // _Pragma unroll
72        // J'aurais pu "roller" les 4 cas des ifs.
73        // Mais pour les perfs, il faudrait tout unroller vu le nombre
74        // d'itérations. Et j'aimerais bien comparer du code unrollé
75        // aux 2 niveaux à la main avec celui unrollé par le compilateur
76        // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
77        // l'intérieur d'abord (pour ne pas refaire le même boulot)
78        // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
79        // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
80        $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
81        $i_rest = intdiv(
82          $i_rest - $i_significant_bits,
83          2**$arr_data["i_bits"],
84        );
85        $s_result = (  // .= or =.
86          chr($i_significant_bits + $arr_data["i_base_value"])
87          .$s_result
88        );
89      }
90      return $s_result;
```

```
91    }
92    // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
93    if($i_code_point_in_decimal_notation < 256 * 256 * (2**5)){
94      $i_first_byte_base_value = 240;
95      $arr_arr_data_per_byte_reverse = [
96        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
97        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
98        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
99        ["i_bits" => 3, "i_base_value" => $i_first_byte_base_value],
100     ];
101     $i_rest = $i_code_point_in_decimal_notation;
102     $s_result = "";
103     foreach($arr_arr_data_per_byte_reverse as $arr_data){
104       // _Pragma unroll
105       $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
106       $i_rest = intdiv(
107         $i_rest - $i_significant_bits,
108         2**$arr_data["i_bits"],
109       );
110       $s_result = (  // .= or =.
111         chr($i_significant_bits + $arr_data["i_base_value"])
112         .$s_result
113       );
114     }
115     return $s_result;
116   }
117   throw new Exception(
118     "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
119     ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
120     ." 3 octets au lieu de 4 :face-palm:."
121     ." Du coup, ça peut donner quelques \"surprises\"."
122     ." Sinon, y'a aussi ASCII dans MySQL."
123     ." Et comme ça joue sur la mémoire et les performances,"
124     ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
125     ." :)"
126   );
127 }
128
129
130
131 function hexa_code_point_to_UTF8(
132   $s_code_point_in_hexadecimal_notation
133 ){
134   $s_code_point_in_hexadecimal_notation = strtoupper(
135     $s_code_point_in_hexadecimal_notation
136   );
```

```
137   $i_code_point_in_decimal_notation = 0;
138   $i_ord_0 = ord('0');
139   $i_ord_A = ord('A');
140   $i_digit_offset = -$i_ord_0;
141   $i_letter_offset = 10 - $i_ord_A;
142   for(
143     $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
144     $i < $i_max;
145     ++$i
146   ){
147     $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
148     $i_code_point_in_decimal_notation *= 16;
149     if(ctype_digit($s_hexa_digit)){
150       $i_code_point_in_decimal_notation += (
151         ord($s_hexa_digit) + $i_digit_offset
152       );
153       continue;
154     }
155     if(ctype_xdigit($s_hexa_digit)){
156       $i_code_point_in_decimal_notation += (
157         ord($s_hexa_digit) + $i_letter_offset
158       );
159       continue;
160     }
161     throw new Exception("Not an hexadecimal digit.");
162   }
163   return decimal_code_point_to_UTF8(
164     $i_code_point_in_decimal_notation
165   );
166 }
167
168
169
170 function check_string_is_valid_ASCII($s_string){
171   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
172     if(ord($s_string[$i]) > 127){
173       throw new Exception(
174         "Non-ASCII character found at octet position "
175         .$i
176         ., value "
177         .ord($s_string[$i])
178         ."."
179       );
180     }
181   }
182   return true;
```

```
183 }
184
185
186
187 function check_file_is_valid_ASCII($s_file_name){
188   $s_string = file_get_contents($s_file_name);
189   if($s_string === false){
190     throw new Exception("File ".$s_file_name." not found.");
191   }
192   return check_string_is_valid_ASCII($s_string);
193 }
194
195
196
197 function check_string_is_valid_UTF8($s_string){
198   $i_continuation_octet_needed = 0;
199   $i_character_start_position = 0;
200   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
201     $i_current_octet = ord($s_string[$i]);
202     if($i_continuation_octet_needed > 0){
203       if($i_current_octet < 128 || $i_current_octet >= 192){
204         throw new Exception(
205           "Non-UTF8 character found at start position "
206           .$i_character_start_position
207           ., octet at position "
208           .$i
209           ." has value "
210           .$i_current_octet
211           ." which is not a continuation octet."
212         );
213       }
214       --$i_continuation_octet_needed;
215       continue;
216     }
217
218     $i_character_start_position = $i;
219     if($i_current_octet < 128){ // 0xxxxxxx ASCII
220       continue;
221     }
222     if($i_current_octet >= 128 && $i_current_octet < 192){
223       throw new Exception(
224         "Non-UTF8 character found at start position "
225         .$i_character_start_position
226         ., octet at position "
227         .$i
228         ." has value "
```

```
229          .$i_current_octet
230          ." which is a continuation octet."
231        );
232      }
233      if($i_current_octet >= 192 && $i_current_octet < 224){
234        // 110xxxxx 10xxxxxx
235        $i_continuation_octet_needed = 1;
236        continue;
237      }
238      if($i_current_octet >= 224 && $i_current_octet < 240){
239        // 1110xxxx 10xxxxxx 10xxxxxx
240        $i_continuation_octet_needed = 2;
241        continue;
242      }
243      if($i_current_octet >= 240 && $i_current_octet < 248){
244        // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
245        $i_continuation_octet_needed = 3;
246        continue;
247      }
248      throw new Exception(
249        "Non-UTF8 character found at start position "
250        .$i_character_start_position
251        ., octet at position "
252        .$i
253        ." has value "
254        .$i_current_octet
255        ." which is invalid."
256      );
257    }
258    return true;
259  }
260
261
262
263  function check_file_is_valid_UTF8($s_file_name){
264    $s_string = file_get_contents($s_file_name);
265    if($s_string === false){
266      throw new Exception("File ".$s_file_name." not found.");
267    }
268    return check_string_is_valid_UTF8($s_string);
269  }
270
271
272
273  /*
274  ?>
```

```
<?php
require_once("./unicode.libr.php");
var_dump(hexa_code_point_to_UTF8("002B"));
var_dump(hexa_code_point_to_UTF8("00E6"));
var_dump(hexa_code_point_to_UTF8("1400"));
var_dump(hexa_code_point_to_UTF8("10111"));
// https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
// Some invisible unicode characters that are a problem:
// 2000-200F 2028-202F 205F 2060-206F
$arr = [
  '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
foreach($arr as $s_hexa){
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  | sed -e 's/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
}
//*/

```

2.56 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
```

```
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# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
  fi
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed_point_build* ]]; then
    LFBFL_fixed_point_build="--fixed_point_build"
  fi
  readonly LFBFL_verbose
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  ./build_and_checks.exec.sh "." "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"
}

upgrade_build_and_checks "$@"
```

2.57 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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#
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# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1 file_path
  # $2 download_URL
  # $3 correct_sha512
  # $4 optional --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 wget_sha512 $*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  if [[ ! -f "$1" ]];
  then
    wget "${LFBFL_verbose}" -O "$1" "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum "$1"\
    | cut -f1 -d' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]];
  then
    echo "$1 does not have correct sha512"
    echo "wanted $3"
```

```
    echo "found ${LFBFL_present_sha512}"
    exit
  fi
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !